

Who Builds in America: Homebuilders and Housing Supply*

Augustus Kmetz[†]

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Abstract

This paper presents new evidence on U.S. housing production before and after the Great Recession. New construction fell 75 percent from peak to trough, and owner-oriented segments have not regained their 2000 levels of production, even in areas with ample land and low regulation. More than half of residential builders exited during the bust, with recovery driven almost entirely by large firms. Small and medium builders remain below their 2000 activity and entry collapsed. Firm size heterogeneity shapes both products and financing: large firms sell standardized, lower-cost homes to mortgage-dependent buyers, while smaller firms produce a variety of products purchased with cash or conventional loans. These patterns indicate that the traditional built-for-sale construction model was persistently disrupted, leaving an industry where housing supply is scarred and large firms dominate new home construction.

1 Introduction

There is a housing shortage in the United States. Since 2012, house prices have boomed, but the supply response has been lower than historical norms. From 2000 to 2006, national house prices rose by 32.7% and the construction sector permitted 9.44 million new single-family homes. From 2013 to 2019, national house prices rose by 29.4%, but only 5,047,287 new single-family homes were permitted. Figure 1 shows that there has been a break from previous housing volume cycles. In the past, the decline was followed by a swift recovery. Instead, the housing bust was deeper and longer than any recorded before. The urban literature typically points to regulatory constraints as the primary factor limiting housing production. However, the housing boom and bust, which was central to the construction and housing sectors, is more plausible as a mechanism to explain the housing shortage.

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[†]Email: guskmetz@gmail.com

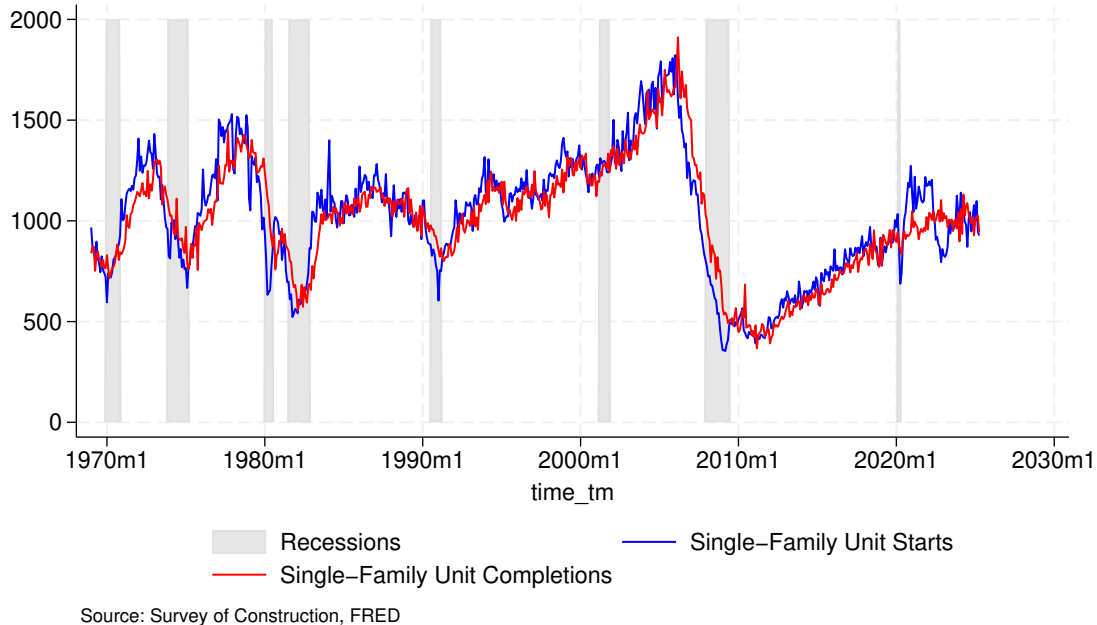


Figure 1: Single-Family Housing Starts and Completions

Notes: This plot shows starts and completions, seasonally-adjusted at an annual rate, for new single-family housing in the United States from 1970 to 2025.

This paper presents new facts about the production of housing in the United States. Aggregate permits fell 75% from peak to trough. None of the owner-oriented segments of home production recovered to their 2000 production levels. In the cross-section, home production fell, even in areas with low regulatory burden and high land availability. On the firm level, the Great Recession wiped out over half of residential builders. The recent boom has been dominated by large builders; small and medium-sized firms remain below their 2000 levels of firm counts and home production. Only the large builders recover to their previous levels of production. This is consistent with anecdotal evidence of credit constraints on the firm side affecting the recovery. These firms operate in different ways, with large firms selling larger, more standardized homes for lower prices per square foot and small firms selling more tailored products. Because of the different ways builders finance their operations and do business, builders have heterogeneous preferences over mortgage sources. Small builders have more buyers using cash and conventional mortgages while large builders rely more on mortgage buyers and FHA loans.

Using data since 1990 at the Core-Based Statistical Area (CBSA) level, I show that aggregate housing production fell and did not recover from the Great Recession in both new permits and completions. The recovery was geographically heterogeneous, but the 2019 distribution of new single-family permits per thousand new households is first-order

stochastically dominated by the 1990 distribution. Defining a recovery as a CBSA returning to its minimal permits rate from 1990-2000, I show that fewer than 40% of CBSAs returned to their pre-crisis permits rate. This lack of recovery is geographically widespread, and is present even in areas with low regulatory burdens.

I use transactions data on newly-built homes from 2000 to 2019 to document facts about the firms who sell homes in the built-for-sale segment. The Great Recession wiped out over half of builders, and while the aggregate quantity of new construction sales has almost recovered to its 2000 levels, the composition of sold homes has changed. Much of this recovery and change in market share is driven by large builders. Large builders account for 61% of new homes in 2019, up from 44% in 2000, and small- and medium-sized firms produce less than 75% of their 2000 levels. At the same time, entry has fallen, even with strong price growth, and the entrant share of new homes has fallen with it. The recovery in construction among large builders is accounted for by the outsized recovery of the largest firms.

Finally, I document how builders do business over time: the size of the builder affects how it operates. Large builders build homes that are larger overall, more similar, and cheaper per square foot than small builder homes. Small builders' customers rely more on cash and conventional mortgages relative to other builders. Large builders' customers are more likely to use mortgages and FHA loans to fund their purchases. When credit eased during the housing boom, builders accepted payment in more conventional mortgages, and when access to those mortgages fell, they accepted more FHA loans.

Since [Dipasquale \(1999\)](#)'s classic note about the lack of academic knowledge about housing supply, a literature has proliferated studying housing supply. [Mayer and Somerville \(2000b\)](#) established that house prices tend to move more in response to changes than to levels. Since then, the housing supply literature has focused on studying how supply responds to changes in price, or *supply elasticities*. A focus has been on the causes and consequences of lower supply elasticities. Barriers such as land unavailability ([Gyourko, Mayer, and Sinai \(2013\)](#), [Saiz \(2010\)](#)) and housing regulation ([Mayer and Somerville \(2000a\)](#), [Gyourko, Saiz, and Summers \(2008\)](#), [Gyourko, Hartley, and Krimmel \(2021\)](#)) along several dimensions affect the response of housing supply to price changes at the metro and within-metro level [Baum-Snow and Han \(2024\)](#). Recent work by [Aastveit, Albuquerque, and Anundsen \(2023\)](#) comparing the two most recent booms finds that the measured supply elasticity has effectively disappeared in all cities in the most recent boom as compared to the previous boom, but that the decline in supply elasticity was worse in areas with higher levels of regulation. [Glaeser and Gyourko \(2025\)](#) provide descriptive evidence that even historically high-production MSAs are now producing relatively less than in the past, and posit that the suburban frontier is closing. A smaller literature has pushed back on the use of supply

elasticities, as they are often correlated with demand (Davidoff, 2013) or explain little about the long-run evolution of housing prices or the response of housing to demand shocks (Louie, Mondragon, and Wieland, 2025). In my analysis, I provide descriptive evidence that the slowdown in supply response has been more widespread than just in highly-regulated areas, and that the Great Recession is the driver.

Macroeconomics, housing, credit, and the business cycle are deeply connected (Piazzesi and Schneider, 2016). Indeed, Leamer (2007) shows that residential fixed investment was a primary force of business cycles; housing followed a volume cycle, not a price cycle. In Yagan (2019), the Great Recession had long-lasting impacts on affected workers and in Le (2025) price drops around the Great Recession lead to persistent slowdowns in construction. A line within this literature has studied the role of credit on outcomes. In Justiniano, Primiceri, and Tambalotti (2019), the housing boom was consistent with a loosening of lending constraints. Chodorow-Reich (2013) finds that small firms with less exposure to crisis-exposed banks in terms of their lending arrangements fared better than firms with more exposure. Mian and Sufi (2016) find that price drops coupled with high levels of household debt can account for the large and persistent declines in consumption. Ayres and Raveendranathan (2023) show that shocks to the credit constraint can produce persistent recessions by reducing small firm entry and hastening small-firm exit. It is difficult to directly measure credit constraints, but I provide evidence of credit tightening for firms in the construction sector. The recent boom is dominated by large firms, and small firms enter less, which is consistent with a credit tightening story.

Finally, this work contributes to a small but growing literature on the industrial organization of construction. Kroft, Luo, Mogstad, and Setzler (2025) finds that construction firms exert monopsony and monopoly power to force wage markdowns of more than 30% and price markups of around 45% in government procurement auctions. In the residential construction context, this is difficult to square with Le (2025) who finds a rapid increase in construction wages. Using public data, Ahluwalia, Baker, and Colton (2022) notes that concentration has risen, and large firms like D.R. Horton and Lennar account for a significant amount of production. Quintero (2022) finds that declines in the number of firms required to reach 90% of the housing production of an MSA, or increased concentration, lead to declines in housing volumes and increases in price volatility. Relatively little is known about the activity and dynamics of firms in the residential construction sector. To my knowledge, the only work directly examining the supply decision-making of residential construction firms is van Straelen (2023), which shows that in times of crisis, large builders mark down homes in less-affected markets to generate free cash flow. I will provide evidence on the market structure of new homes and how residential construction firms operate.

Section 2 provides institutional details for the homebuilding industry. Section 3 discusses data construction. Section 4 lays out new facts on aggregate and cross-sectional home production. Section 5 describes the increasing role of large firms in the post-Recession period. Section 6 shows some of the characteristics of homes built by firms in different segments. Section 7 concludes.

2 Institutional Setting

In 2019, residential fixed investment was 3.8% of GDP. In 2000, it was 4.7% of GDP. Single-family housing is the majority of new housing produced in the United States. While the single-family share of total permitted units fell from 74% in 2000 to 59% in 2019, single-family housing remains the largest source of new units and new buildings.¹ The residential building construction sector is skewed in terms of firm size. In Census data from 2000, there were 133,478 firms operating, and of these, 128,916 had fewer than 20 employees (96.58%). In 2019, my final year of study, the industry remained fragmented. There were 149,190 total firms, with 144,665 firms having 20 or fewer employees (96.97%).² A caveat is that not all of these are builders. Builders are those who organize and manage the construction of the home. Many of these firms are sub-contractors who specialize in certain parts of the building process such as plumbing, electrical work, insulation, framing, and roofing. Construction volume is similarly skewed. In my dataset of new construction, the largest 100 builders account for over half of new homes by 2019.

To build a house, a builder needs land, labor, materials, and financing to pay for the first three. In 2019, according to the 2019 NAHB Cost of Constructing a Home Survey, the average total sales price of a new single-family home was \$485,128. The finished lot cost was \$89,540, (18.5% of total price) and the total construction cost (which includes labor and materials) was \$296,652 (61.1% of total price). The typical builder profit was \$44,092 (9.1% of total price) after overhead and other expenses. Construction costs are the largest share of expenses in the typical unit. Nominal construction costs rose 42.5% from \$80 per square foot in 2011 to \$114 dollars per square foot in 2019.

There are two main market segments: built-for-sale homes and custom homes. A built-for-sale home³ is a home built before a buyer is found.⁴ Builders in this segment are often called *speculative builders*. As shown in Figure 2, the leading source of single-family homes

¹Unit share is calculated as the number of single-unit homes divided by the total number of units permitted from the Building Permits Survey.

²Statistics calculated using [Business Dynamics Statistics Explorer](#) for NAICS code 2361.

³Often called a “spec” home by industry participants.

⁴See [Census Survey of Construction Definitions](#).

is the built-for-sale segment. This is in contrast to the custom-built or custom market where the customer will specify the details of the house to be constructed, or the builder will design a house in consultation with the customer. General contractors organize the construction of the house. An owner-built house is one in which the owner of the land acts as general contractor, and a contractor-built house is one where a paid builder acts as general contractor. A small (but growing) segment is the built-for-rent segment, in which a builder or other entity builds a single-family home in order to rent it out.

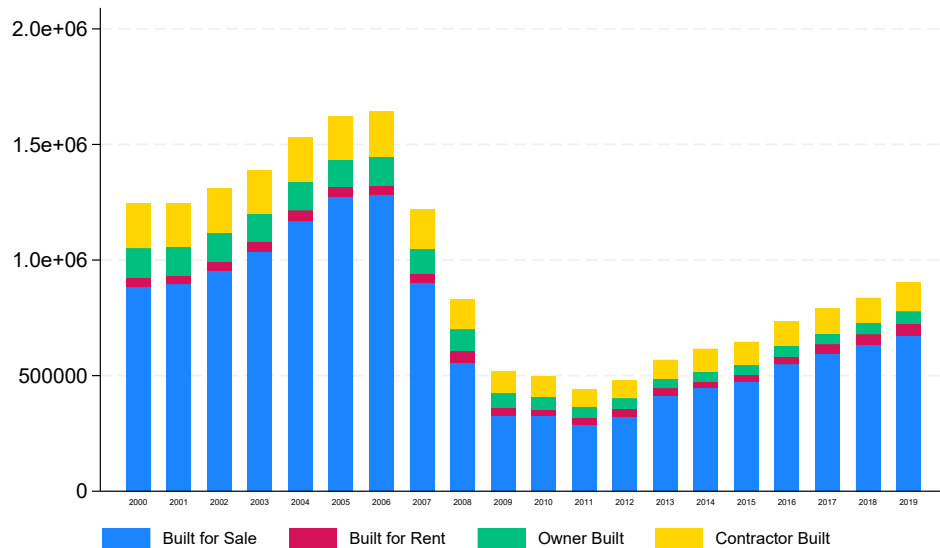


Figure 2: Survey of Construction Counts of Construction Purpose

Notes: This plot shows the number of homes completed in a calendar year from the Survey of Construction by market segment. Sums for built-for-rent and built-for-sale were interpolated in 2017 and 2018 because the categories were merged in those years

The leading homebuilding trade association is the National Association of Home Builders (NAHB). The NAHB defines small builders as those who build 25 or fewer homes per year, medium builders as those who build 26-100 homes per year, and large builders as those who build more than 100 homes per year (Allen, Schreyer, and Thallon, 2022). There is significant heterogeneity in the operations of firms based on size. When engaged in building as contractors, small builders typically build large custom homes, where the overall project cost and margins are higher. Some small builders also do speculative building. Medium and large builders usually act as speculative builders and build less-expensive similar homes on large tracts of land that they develop to benefit from economies of scale (Allen, Schreyer, and Thallon, 2022). Most large builder homes within a location follow a set of home plans for that location. This enables mass production. Large builders also typically have a lower cost of capital, and lower build costs because large builders can negotiate bulk labor and

materials rates. Smaller firms in the sector may prefer to work as sub-contractors for these medium and large builders because they avoid the direct financial risk of speculative building and sub-contracting for large builders can offer consistent employment.

Credit is critical for construction. Credit plays a role at both the beginning and end of building a house. For custom homes, only the buyer needs to secure financing. The buyer either pays cash or procures a construction loan and a mortgage for when construction finishes. The construction loan can be bundled with the mortgage in a construction-to-permanent loan to reduce the administrative costs by underwriting one loan instead of two. For speculative homes, the buyer and builder must both secure financing. For the buyer, this will take the form of cash or a mortgage loan. One advantage of large builders is that they often have internal mortgage units, which allow them to originate mortgages internally and then sell them off for securitization. For the speculative builder, financing is needed to start construction. In Q1 2005, the NAHB Survey on Acquisition, Development & Construction Financing reported that 92% of respondents sought new loans; this fell to below 50% since the housing bust.

The financing strategies of speculative builders vary with their size. Small builders assume significant risk when building speculative homes. Built-for-sale homes produced by small builders are typically financed using a construction loan from a bank. A construction loan is a short-term loan intended to finance the construction of a specific building. The builder purchases the lot beforehand, and the land serves as collateral for the loan. Sometimes, banks will require firms to use their entire portfolio as collateral for the loan. The term for single-property construction loans usually varies from 12 to 15 months ([Allen, Schreyer, and Thallon, 2022](#)). The mean speculatively-built single-family home loan term was 12.5 months in Q2 2025 according to the National Association of Home Builders.⁵ The loan accumulates interest based on the expenses accrued, similar to a line of credit, but the loan is tied to the specific property. At the end of the construction loan, the small builder must pay back the loan or convert it into a mortgage. While builders can request extensions, they are exposed to timing risk because interest rates are generally higher than mortgage rates and builders may lack other cashflows to pay back the loan at maturity if the house does not sell in time. The objective of the small builder is then to sell the home as quickly and profitably as possible.

Large builders have more flexibility in their financing options, but still rely on a combination of credit and cash. Large builders have lines of credit or other credit arrangements to finance their new builds. These arrangements are not usually tied to specific properties, but rather the portfolio in aggregate. In 2019, D.R. Horton maintained an unsecured revolving

⁵See Exhibit 4 at [NAHB's Acquisition, Development and Construction Financing Survey](#)

credit facility and issued senior homebuilding notes to fund homebuilding operations.⁶ However, there are usually covenants or other restrictions involved with these lines of credit, so firms may have to rely on cash if their collateral value deteriorates.

During the 2008 crash and subsequent recession, credit tightened for firms and their customers. The NAHB Survey on Acquisition, Development & Construction Financing reported that its respondents faced tightening credit conditions from 2006 to 2012.⁷, with conditions only starting to ease somewhat by 2013. During the crisis, because the value of their collateral (the land and properties they held) had fallen so much, Lennar and D.R. Horton could not draw on their lines of credit because of financial covenants in the lines of credit. [van Straelen \(2023\)](#) studies how these large house sellers generated cash flow by marking down properties in less-affected markets in order to survive. Anecdotal evidence from industry participants suggests that credit constraints mattered to firms after the bust. In April 2012, Donald J. Tomnitz, the CEO of D.R. Horton noted in *Builder* magazine that “this time around, the private builders have been devastated more than they have been devastated in any previous downturn ... So I see our markets and our sub-markets around the country, and the one thing that is consistent among all these markets is that there are fewer and fewer private builders in there who are [still] alive, and if they are alive, [they] can’t get a loan from a bank.” There is reason to believe that credit matters, especially for speculative builders, and that credit availability tightened after the crash. After the crash, there was increased merger activity, particularly among the larger home builders ([Quintero, 2022](#)).

3 Data

I build two datasets to examine housing supply in the United States. To analyze housing production over time, I construct a dataset at the CBSA level that contains information on 869 different CBSAs from 1990 to 2019. To measure the firm structure of the U.S. built-for-sale housing market, I construct a dataset using data sourced from the Cotality Owner Transfer and Mortgage datasets of newly-constructed homes in the United States across 165 CBSAs.

3.1 Panel Data Construction

I construct an annual panel data set following 869 US CBSAs from 1990 to 2019. I collect data on new permits, population, population inflows, housing prices, regulatory burdens, and

⁶Page 44 of [2024 Annual Report](#).

⁷Page 5 of [NAHB AD&C Survey Report](#)

financing sources. To measure permits, I use the county-level Building Permits Survey (BPS). The BPS is a representative monthly panel of housing production by unit size, available at the county, MSA, state, and national level. I use the NBER CBSA-County crosswalk⁸ to aggregate permits from the county to the CBSA level. I focus on single-family permits.

To measure population, I use the National Cancer Institute Surveillance, Epidemiology, and End Results (SEER) Program U.S. County Population Data, which tabulate county population by age. I take total population by county and aggregate to CBSA-level using the NBER CBSA-County crosswalk. To measure population inflows, I use the Internal Revenue Service Statistics of Income Division (SOI) inflow migration data, which tabulate the number of households and people who moved into a county from another destination. I aggregate total inflows from the county-level to the CBSA-level and adjust for internal migration using the source county information.

To measure home prices at the CBSA level, I use the FHFA’s Home Price Indices. These are constructed using sale and appraisal data and are generally available since 1980. I match using the CBSA code.⁹ I then adjust these into real terms using CPI inflation excluding shelter.¹⁰

I merge these aggregated data on CBSA code. Table 1 shows the number of CBSAs remaining after each merge, as well as the initial share of the population represented.

Merge step	CBSAs before	CBSAs matched	CBSAs after	Population share (%)
Initial permits data	917		917	
SEER population	917	917	917	100.0
SOI migration	917	917	917	99.9
FHFA HPI	917	869	869	98.5
FICO credit data	869	865	869	97.2
Cotality transactions	869	165	869	77.6
Saiz regulation data	869	251	869	81.5
Final panel	869		869	

Table 1: Coverage of U.S. CBSAs in Panel Data

Notes: Data come from the Building Permits Survey, National Cancer Institute SEER U.S. County Population Data, SOI Migration Statistics, FHFA CBSA House Price Indices, aggregated FICO scores by CBSA from Cotality Loan-Level Market Analytics, aggregated transactions data from Cotality Owner Transfer and Mortgage, and Saiz (2010). Each row reports the number of CBSAs before merging, matched in the merge, and remaining after merging in additional datasets, along with the share of national population represented by each sub-merge. I only keep CBSAs that I match to the permits, population, SOI, and FHFA HPI data. I do not require matches for the FICO data, Cotality transactions data, and the Saiz data, as these cover smaller subsets of the data.

⁸Sourced from [NBER CBSA to FIPS County Crosswalk](#)

⁹I perform adjustments to fix incorrect CBSA codes for Cleveland, OH, Lexington Park, MD, and The Villages, FL.

¹⁰I use the series CUSR0000SA0L2 from [FRED](#).

3.2 Built-for-Sale Data Construction

To measure the built-for-sale segment, I construct a dataset of new-build homes using the Cotality Owner Transfer and Mortgage datasets. I use the Owner Transfer dataset as the base data. The Owner Transfer data contain detailed information on transactions, including the type, participants, timing, amount, and whether the transactions used cash or a mortgage. Since real estate transactions are public records available from county assessor offices, these data are generally accurate. I limit to single-family homes sold in arms-length transactions. To get information on the source of financing, I augment my data with the Cotality Mortgage data to observe more details about the mortgage origination. I construct an identifier that maps a property to the primary mortgage on the transaction. I then merge the two datasets. I keep all homes sold in 1990 or later that are marked with the Cotality proprietary "New Construction" indicator.

After these basic steps, I have a dataset of 13,224,437 observations. Table 2 details the subsequent cleaning steps. I drop interfamily transactions. I construct a unique ID using the APN, APN Sequence Number, and FIPS code and keep only the first transaction to get the first sale date of a new construction. I drop purchases where we do not observe the source of funds and mortgage-funded transactions where we do not observe the mortgage. I drop custom homes and only keep CBSAs where there are more than 10 transactions each year from 2000 to 2019. I also perform basic cleaning of the firm names in order to ensure we can measure and match firms over time.

My final data contain 3,757,459 observations across 165 CBSAs from 2000 to 2019. The correlation between aggregate permits from the Build Permits Survey and single-family homes sold in the data is 0.9525, indicating that the aggregate series track each other closely. Table 3 shows that the 165 CBSAs represent a majority of the United States' population and new permits. The CBSAs I observe contain 78% of all permits issued and 77% of the population from 2000 to 2019. I observe new construction equivalent to 28% of all single-family permits issued within the matched CBSAs.

Step Description	Observations Remaining	CBSAs
Import raw CoreLogic data	13,224,437	
Drop interfamily-related transactions	13,195,362	
Keep earliest sale per parcel (UNIQUE_ID)	6,133,562	
Drop “Other” purchases	5,981,414	
Drop mortgages with missing loan info	5,981,120	
Merge with CBSA crosswalk	5,892,207	830
Drop sellers with “CUSTOM” in name	5,888,263	830
Restrict to CBSAs with full panel coverage (2000–2019)	5,190,828	165
Drop duplicate transactions (unique_id, seller, buyer, sale amount)	5,181,338	165
Keep only a single occurrence of UNIQUE_ID	5,172,896	165
Restrict to years 2000–2019	3,757,459	165

Table 2: Cleaning steps applied to the Cotality (CoreLogic) dataset. The final cleaned dataset contains 3,757,459 newly constructed, arms-length, single-family residential properties (2000–2019), drawn from 165 CBSAs with full panel coverage, suitable for analysis of the build-to-sell market.

Measure	All CBSAs	Matched CBSAs	Share Matched
Number of CBSA-year observations (permits data)	18,268	3296	0.18
Single-family permits issued	17,677,887	13,391,467	0.76
Population (000s)	5,703,996	4,211,303	0.74
CoreLogic transactions vs. permits in matched CBSAs	13,391,467	3,754,446	0.28

Table 3: Representativeness of New Construction Data. The matched CBSAs account for about 18% of CBSA-year observations, but cover 76% of single-family permits issued and 74% of the U.S. metropolitan population. Within these matched CBSAs, Cotality transactions represent approximately 28% of all permits issued. Data on permits come from the U.S. Census Bureau Building Permits Survey, and population data come from the National Cancer Institute Surveillance, Epidemiology, and End Results (SEER) Program U.S. County Population Data.

4 Housing Production Fell and Did Not Recover

During the housing boom, prices fell 31% from the peak in 2006 to the low in 2012. However, by 2015, housing prices had returned to their 2000 levels, and price growth was stronger than 2000 levels. Using data from the BPS, Figure 3 shows that new single-family permits peaked in 2005, at 1.6 million new single-family homes permitted before falling 75% to 396,910 new units in 2011. New single-family permits had still only recovered to 862,084 units by 2019, which is 28% lower than its 2000 total.

The Census Bureau Survey of Construction (SOC) tracks single-family units under construction in the United States and produces annual microdata. These microdata include detailed information on the timing and attributes of the construction of new single-family buildings in the United States, but do not contain CBSA-level geographical identifiers. Mirroring the decline in permits, Figure 4 shows that completions from the Survey of Construction fell from 1,643,124 completions in 2000 to 442,322 in 2011, a 73% drop. Completions were 902,758 in 2019. The SOC contains information on the purpose of the construction. In the SOC, new single-family homes can be built-for-sale, owner-built, contractor-built, or built-for-rent. Figure 5 shows the number of completions relative to 2000 levels of construction for the different segments in the single-family home market.¹¹

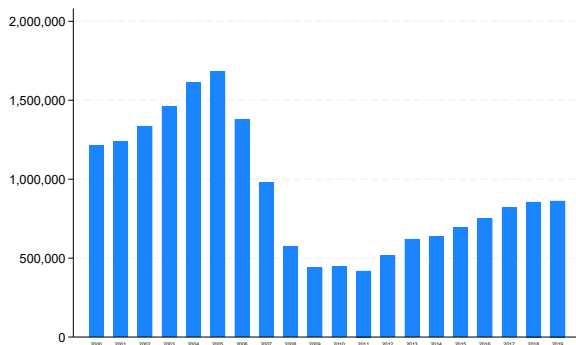


Figure 3: Started Single-Family Units in Building Permits Survey (2000-2019)
Notes: The plot shows the number of started single-family units per year from the Building Permits Survey.

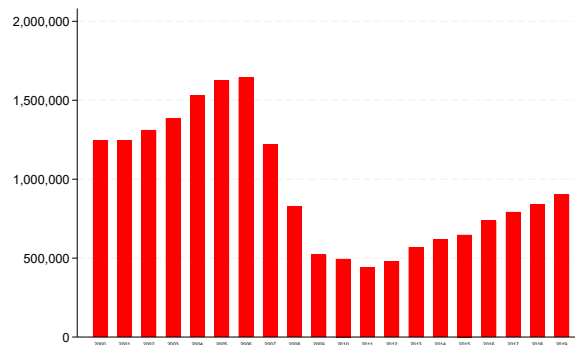


Figure 4: Completed Single-Family Units in Survey of Construction (2000-2019)
Notes: The plot shows the number of single-family units completed per year from the Survey of Construction.

However, there is significant geographic heterogeneity. To quantify the cross-sectional disruption, I construct a measure of how many new single-family homes are permitted for every thousand new potential households in a CBSA. I use new single-family permits from the

¹¹Table 7 in the Appendix tabulates the recoveries for each segment.

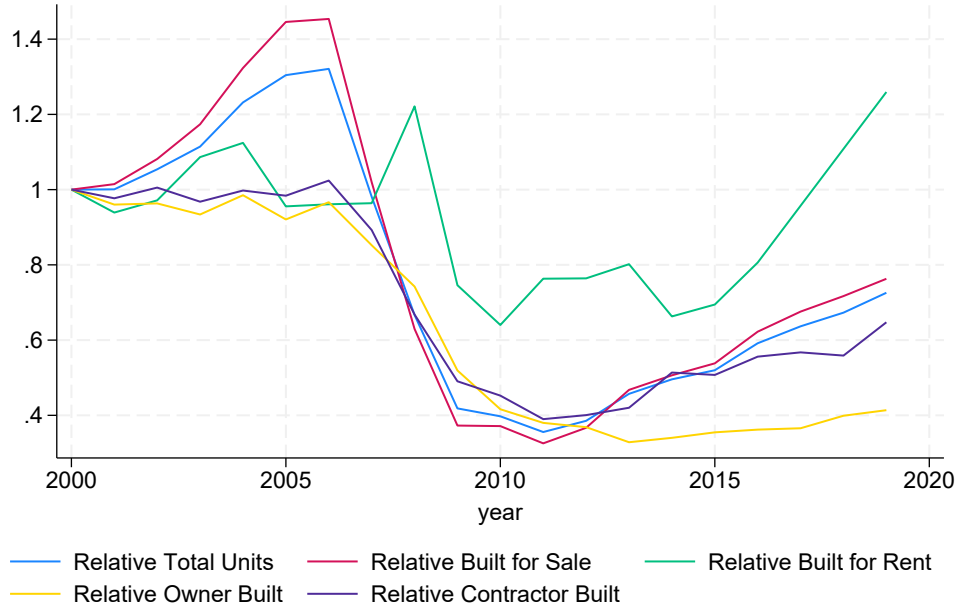


Figure 5: Survey of Construction Relative Recovery of Market Segments
Notes: The plot shows the number of new units produced in each market segment of the Survey of Construction normalized to their 2000 levels.

BPS, and population inflows in thousand returns into a CBSA from the IRS SOI data.¹² Since families typically file a single return, this measures the number of potential new households in a CBSA.¹³ Going forward, I refer to this as the permits rate. I use this measure instead of the number of permits issued because a high level of permit issuance might hide a low rate of permit issuance. If the population growth of a city quadruples over the time period, and the level of permits only rises to 1.1 times the pre-period level of permits, then there will likely be a housing shortage. People who prefer single-family homes will then be forced into apartments, because there are relatively fewer houses available.

Figure 6 shows the distribution of permits rate. In the Appendix ?? shows this in a tabular form. In 1990, the median permits rate is 147.3. Near the start of the boom in 2000, the median is 183.6. After the crash, the entire distribution shifts down. By 2019, the median permits rate only recovers to 94.2. Even though the broader economy recovered from the Great Recession, the distribution of housing production in 1990 first-order stochastically dominates the distribution of production in 2019 (i.e. the 1990 distribution is associated with higher production than the 2019 distribution). This fact points to potential scarring in housing supply. This is consistent with evidence from [Aastveit, Albuquerque, and Anundsen](#)

¹²The SOI Migration data are at a county level, so only inflows from non-CBSA counties are counted new households.

¹³The SOI data also report the number of exemptions, which maps to the number of people moving.

(2023) that supply elasticities are much lower than they were before the crash.

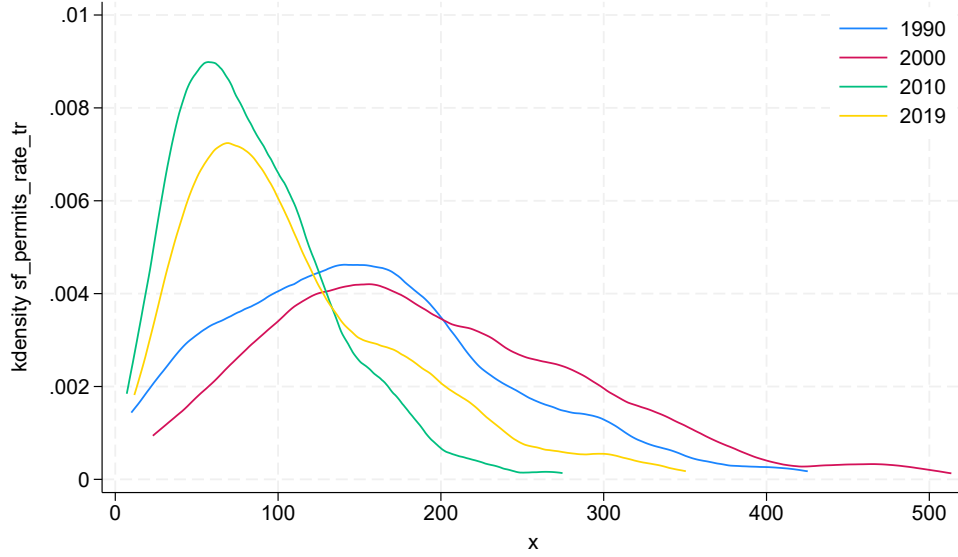


Figure 6: Distribution of Permits per Thousand New Households

Notes: This plot shows the kernel density distribution of new single-family home production divided by the number of new households in thousands for the years 1990, 2000, 2010, and 2019. Each distribution is trimmed at the 1st and 99th percentiles within that year. Permits data come from the Building Permits Survey and inflow data come from the SOI Migration Statistics.

To understand the extent of this scarring, I define the housing supply recovery ratio as the ratio of the CBSA's current permits rate to the lowest permits rate from 1990-2000. According to this measure, Table 4 shows that fewer than 40% of CBSAs have rebounded to their lowest level of production during the 1990s. When CBSAs are weighted by population, fewer than 30% of people live in a CBSA that has recovered in terms of permits rate.

	Mean
year	
2000	1
2005	.9452055
2010	.2694064
2015	.3348554
2019	.3835616

(a) CBSA Recovery (Unweighted)

	Mean
year	
2000	1
2005	.8659624
2010	.1365737
2015	.2213742
2019	.2825391

(b) CBSA Recovery (Population-Weighted)

Table 4: CBSA Recovery of Permits: Comparison of Unweighted vs. Population-Weighted
Notes: These tables compute the share of CBSAs in the data that match or exceed their lowest level of 1990s single-family permits rate (new single-family permits per thousand new households). In the first table, all CBSAs are equally weighted, so this represents the share of CBSAs that recovered. The table on the right is weighted by population, so it instead gives us the share of people who live in a CBSA that has recovered in permits rate.

Figure 7 maps the distribution of housing supply recovery ratios in 2019. Warmer colors are associated with lower permits rates while cooler colors are associated with recoveries. The West Coast and Northeast produce new units at intensities far below those observed in the 1990s, which is consistent with known patterns of land availability and land-use regulation [Saiz \(2010\)](#). However, this pattern is present throughout the Midwest and Great Plains, even in cities like Kansas City and St Louis where there is low regulation and plenty of space to build. There is no clear regulatory or land availability reason for why this scarring is present in so many different CBSAs.¹⁴ Some of these places, such as Atlanta, another low-regulation and high-production city, did not recover to 1990s levels of permits rate, even if it recovered in terms of permits because of having a large amount of population growth.¹⁵ This is consistent with evidence about slowing production by historically high-production areas in [Glaeser and Gyourko \(2025\)](#). While they suggest that this could be due to the closing of the suburban frontier, the housing bust and its consequences are more plausible candidates for such a widespread phenomenon.

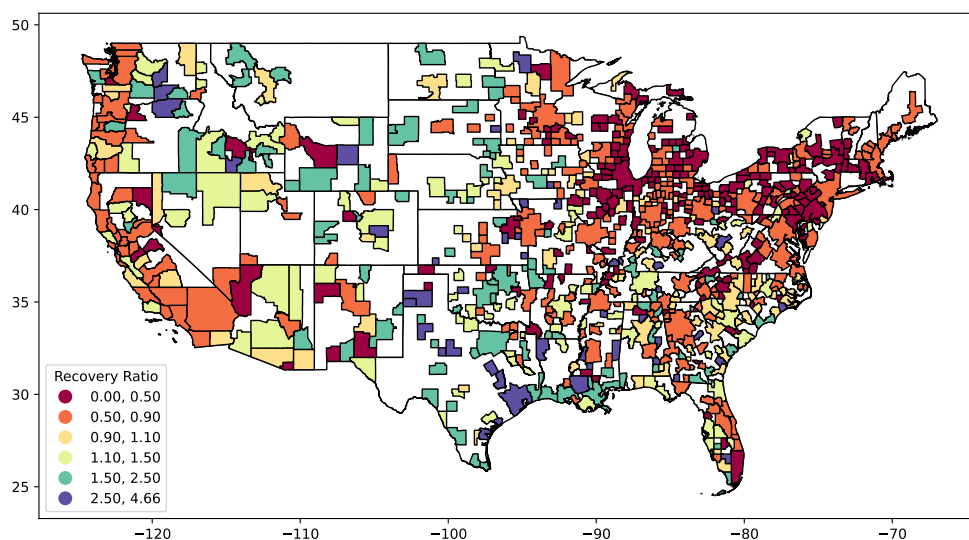


Figure 7: Ratio of Single-Family Permits/Thousand New Households Relative to Minimal 1990s Levels

Notes: This plot is a heat-map of the ratio of 2019 housing permits rate (new single-family permits per thousand new households) to the lowest level of permits rate during the 1990s.

¹⁴See Figure 34, Figure 35, Figure 36, Figure 37, Figure 38, and Figure 39 in the Appendix for the permits rate and relative permits rate time series plots for historically high-production areas, low-production areas, and a selection of cities that house Federal Reserve Banks.

¹⁵Table 8 and Figure 30 in the Appendix show the aggregate figures for the level of permits. Figure 31, Figure 32, and Figure 33 show time-series of high-production, low-production, and Federal Reserve Bank CBSAs.

5 The Built-For-Sale Market is Increasingly Dominated by Large Builders

I now turn to the built-for-sale data. Figure 8 shows the total number of single-family homes sold in that year of data and Figure 9 shows the number of firms selling homes in each year. In 2000, there were 28,996 builders, who sold 194,505 new single-family homes. At the peak in 2005, there were 38,703 active builders. At the trough, this had fallen to 13,545 builders. By 2019, the number of builders had recovered to 21,026. They sold 192,800 single-family homes.¹⁶

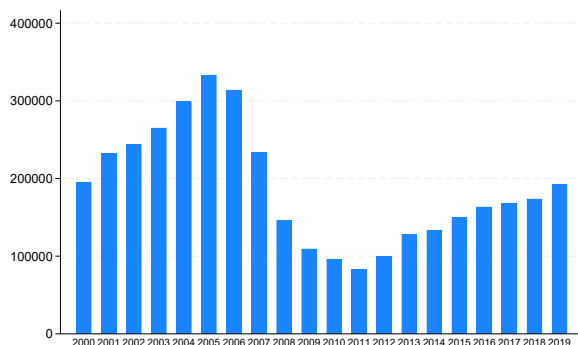


Figure 8: Total Number of Homes Over Time
Notes: This plot shows the total number of homes over time in the Cotality built-for-sale data set.

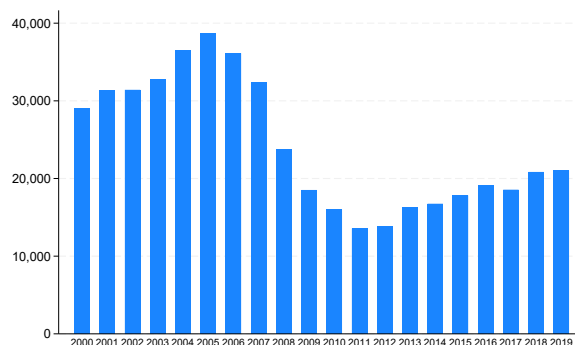


Figure 9: Total Number of Firms Over Time
Notes: This plot shows the total number of firms over time in the Cotality built-for-sale data set.

To understand housing production, we need to understand the market structure. A combination of small, medium, and large builders coexist in the housing market to produce new single-family units. The distribution of builder size is highly skewed. I classify builders according to their annual home production using the NAHB's size categories. Throughout the sample, approximately 97% of builders are small builders, who produce 25 or fewer houses per year. Around 2% are medium-sized builders, who produce 26 to 100 homes, and large builders, who produce more than 100 homes per year, make up less than a percent of all builders. The share of large builders has increased from .6 to .9% of firms. Table 13 in the Appendix shows the annual share of all firms that fall within each NAHB category. Table 10, Table 11, Table 12 in the Appendix show summary statistics for builders in each segment

The production of new housing is similarly skewed. Table 5 shows each size category's annual share of homes sold in each year. In 2000, large firms produce 44.4% of output.

¹⁶The relative recovery of the aggregates may differ from the earlier aggregate figures due to increasing data quality, my sample being only 165 CBSAs, and that the data contain the date of sale and not the date of construction.

By 2019 large firms produce 60.9% of output, a 16.5% increase. Less than 1% of builders produce over half of output. The small builder share of output falls from 40% to 27.3%, and the medium builder share falls from 15.6% to 11.8% of output. This trend agrees with [Ahluwalia, Baker, and Colton \(2022\)](#), who find that the largest 100 builders account for over half of construction nationally. They note the trend has been driven by the two largest firms, D.R. Horton and Lennar. In my data, the increased market share of large firms remains even after removing D.R. Horton and Lennar. In Table 14 in the Appendix, which excludes D.R. Horton and Lennar, the market share of large firms increases by 12.3% to 53.2% in 2019 from 40.9% in 2000.

	Small Builder (1-25 homes)	Medium Builder (26-100 homes)	Large Builder (>100 homes)
year			
2000	.4002056	.1557338	.4440606
2001	.3671387	.1615358	.4713255
2002	.352484	.1615925	.4859236
2003	.3426486	.1620379	.4953134
2004	.3416014	.1545741	.5038245
2005	.3238522	.1523694	.5237784
2006	.3144607	.1405078	.5450315
2007	.3562761	.1312666	.5124573
2008	.3941961	.1204606	.4853433
2009	.410828	.1220913	.4670807
2010	.4066845	.1349709	.4583446
2011	.3954678	.1356075	.4689248
2012	.3625359	.14631	.4911541
2013	.3410398	.1524182	.5065419
2014	.3450852	.1419955	.5129193
2015	.3254344	.1294298	.5451359
2016	.31321	.1273504	.5594396
2017	.288243	.1265966	.5851604
2018	.2889719	.1153884	.5956397
2019	.2728216	.1179668	.6092116

Table 5: Share of Total Homes Built by NAHB Size Category

Notes: This table shows the share of homes built by firms within different NAHB size categories from the Cotality built-for-sale data set.

One contributor for the increased market share of new builders is that the large builders recover to or exceed their 2000 construction levels. The other segments do not. For each year and each size category, I compute the total number of homes sold by firms. I normalize each segment's value so that the sum of homes in each size category is equal to 1 in 2000. Figure 10 shows that large builders sell homes equal to around 136% of their 2000 output in 2019, while medium and small builders produce 75.1% and 67.6% of their 2000 output in 2019.¹⁷

¹⁷Figure 43 shows that this fact is robust to the exclusion of D.R. Horton and Lennar. The relative count is 110% of its 2000 total as opposed to 136%.

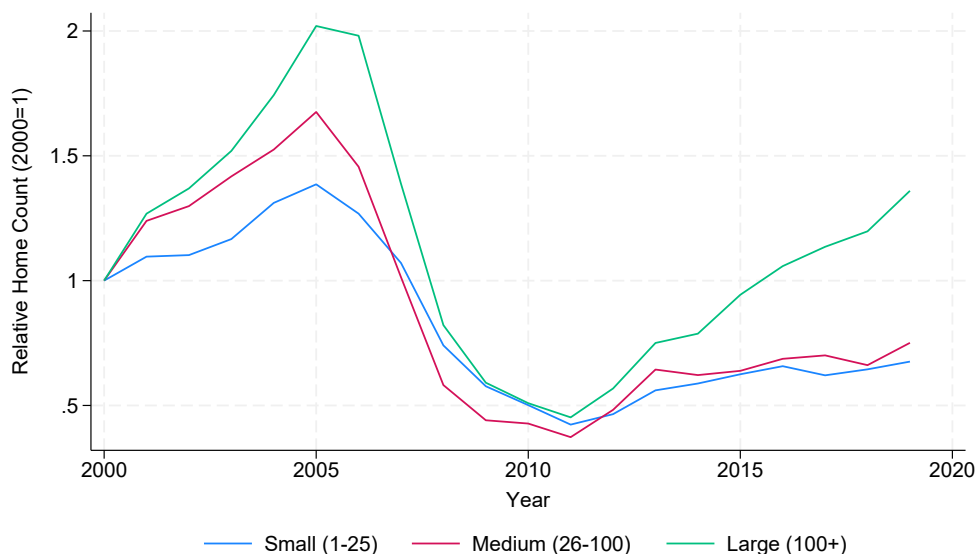


Figure 10: Relative Home Production by Size Category (2000=1)

Notes: This plot shows the relative number of homes produced by each NAHB size category from the Cotality built-for-sale data set. The sum is normalized to be 1 in 2000.

There are differences in the firm recovery, with a bias toward the largest firm, even within the large builders. Figure 11 and Figure 12 show the market share for single-family homes of the ten and hundred largest producers in each year. From 2000 to 2019, the large builders' total market share grew by 16.5%. The market share of the 100 largest builders grew around 17.2% from 38.3% to 55.5%. The market share of the ten largest builders grew 14.7% from 17.3% to 32%. The market share gains disproportionately accrue to the largest firms in the segment. As figures Figure 13 Figure 14 show, the counts for each are much larger than their 2000 output levels and they seem to be much more responsive to price changes. Many of these firms are publicly-traded, which could give them access to deeper and cheaper pools of liquidity than their non-publicly-traded counterparts. They are also more likely to survive. All of the firms in the top 10 in 2000 survive the entire period, and over half of firms in the top 100 in 2000 survive to the end of the period. Figure 44 and Figure 45 in the Appendix show firm survival over time for the Top 10 and Top 100 firms.

While the number of homes sold has almost returned to 2000 levels for large firms, the number of firms operating has not returned to 2000 levels in any size category. I calculate the number of firms operating within each size category and normalize the number of active firms to be 1 in 2000. In Figure 15, the relative number of large and medium firms track together during the boom and subsequent bust, but starting in 2015, they diverge, and the large firm segment reaches 93.6% of its 2000 firm count. The medium and small firm counts are at 73.7% and 72.3% of their 2000 levels respectively. The small firm comparison

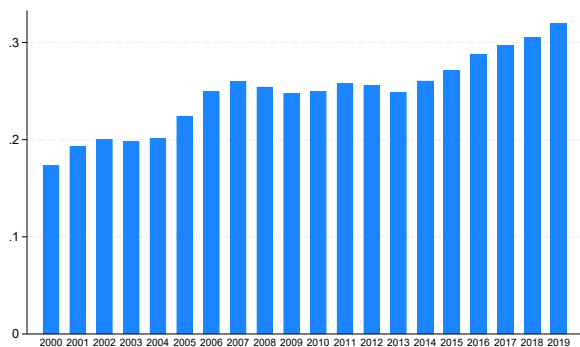


Figure 11: Top 10 Home Share
Notes: This plot shows the share of homes sold by the 10 largest builders active within a calendar year.

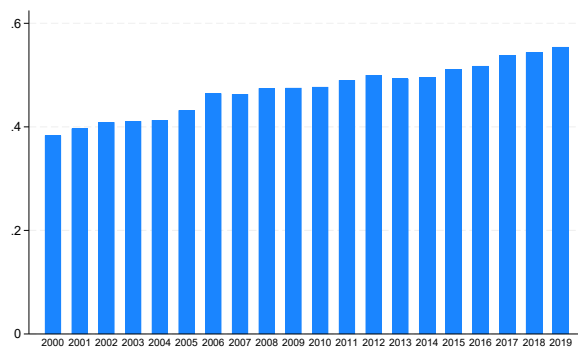


Figure 12: Top 100 Home Share
Notes: This plot shows the share of homes sold by the 100 largest builders active within a calendar year.

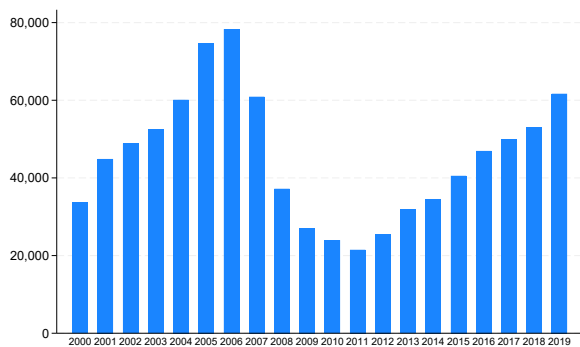


Figure 13: Top 10 Home Counts
Notes: This plot shows the number of homes sold by the 10 largest builders active within a calendar year.

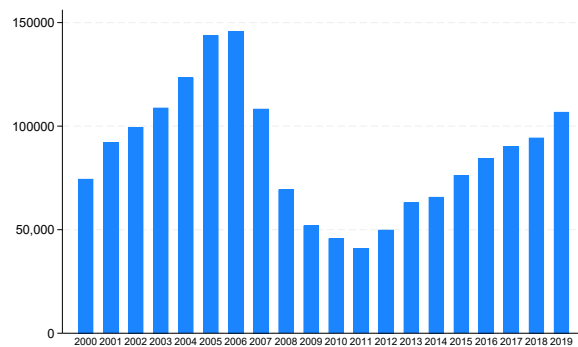


Figure 14: Top 100 Home Counts
Notes: This plot shows the number of homes sold by the 100 largest builders active within a calendar year.

is particularly puzzling. Small builders at 72.3% of the 2000 number of firms produce 67.6% of the 2000 level of small-firm output, which suggests that the small firms active in 2019 are operating at a smaller scale than they did in 2000. On the other hand, the large builder segment is operating on a larger scale than it was in 2000; with fewer firms they produce more output. This may be because there are fewer firms to compete with. There was significant industry consolidation after the crash: D.R. Horton and Lennar (along with other large firms) acquired regional builders or merged with other large builders (Quintero, 2022).

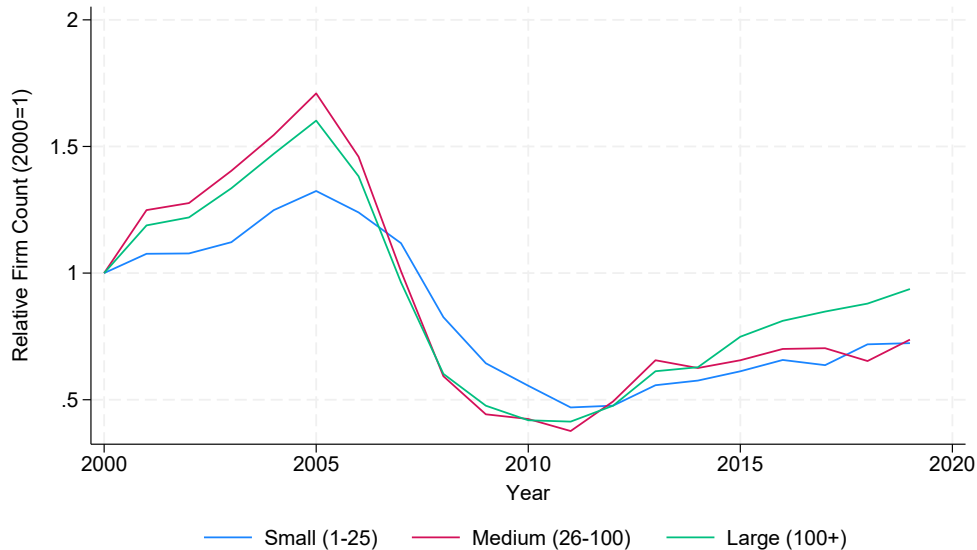


Figure 15: Relative Firm Counts by Size Category (2000=1)

Notes: This plot shows the number of firms present in an NAHB category over time from the Cotality built-for-sale data set. The sum is normalized to 1 in 2000.

Lower entry contributes to persistent low firm counts. After the recession, there are fewer entrants and they produce fewer homes per entrant. To measure firm flows, for each firm name I calculate the first and last years a firm is present in the unfiltered 1990-2024 sample. I define a builder as an entrant or exit depending on if that year is the last year the firm appears in the data.¹⁸ Within entrants, there are two main categories. I define a firm as one-and-done if its first year of operation is its last. Often a new firm will incorporate with the name of an address to build at that address. Once the associated home(s) sell, the company exits. A continuing entrant is a firm that enters but does not exit in its first year.

Figure 16 shows the number of entrant builders in each year. The number of entrants fell from 15,768 before the recession to around 4,429 in the trough, before recovering to 8,964 in 2019. Entry has not returned to its 2000 level. There is also some weak evidence that

¹⁸This way I do not define all firms in 2000 as entrants or all firms in 2019 as exits, since many of the firms existed or continue to exist before and after those years.

the scale of entrants may have fallen. Figure 17 shows that over time, the average number of homes that entrants produce has fallen somewhat from around 3 homes to around 2.6 homes. As a consequence, the share of homes built by entrants has fallen. In the Appendix, Figure 47 shows the declining share of total homes built by entrants over time. The number of continuing entrants fell by over half and has remained low. Figure 18 shows the persistently low number of continuing entrants. There were 6,846 continuing entrants in 2000, 1,732 in 2011, and only 2,912 in 2019. Given that price growth was solid throughout the period, the lack of entry could imply an increased barrier to market entry. Credit tightening is a plausible explanation. Banks may tighten credit in a number of ways: by reducing the number of loans they offer, the amounts they are willing to lend, and the maximum loan-to-value ratio. They may also increase interest rates. In the NAHB Survey on Acquisition, Development, & Construction Financing, these are all listed as common issues builders reported facing when seeking additional loans. These factors could affect builder entry on the extensive and intensive margins.

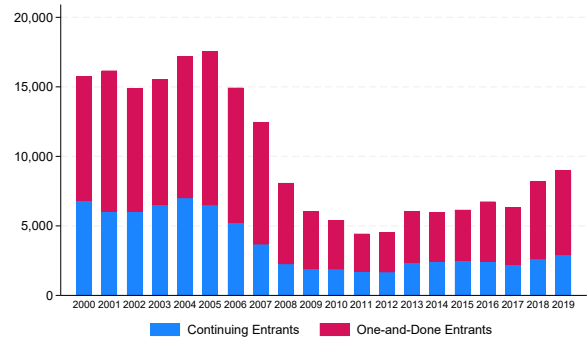


Figure 16: Number of Entrant Builders Over Time

Notes: This plot shows the number of entrant firms in each year of the Cotality built-for-sale data set.

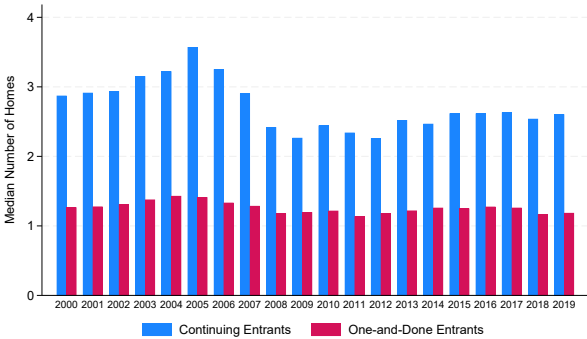


Figure 17: Average Number of Homes per Entrant Builder Over Time

Notes: This plot shows the average number of homes built by entrants by continuation status in the Cotality built-for-sale data set.

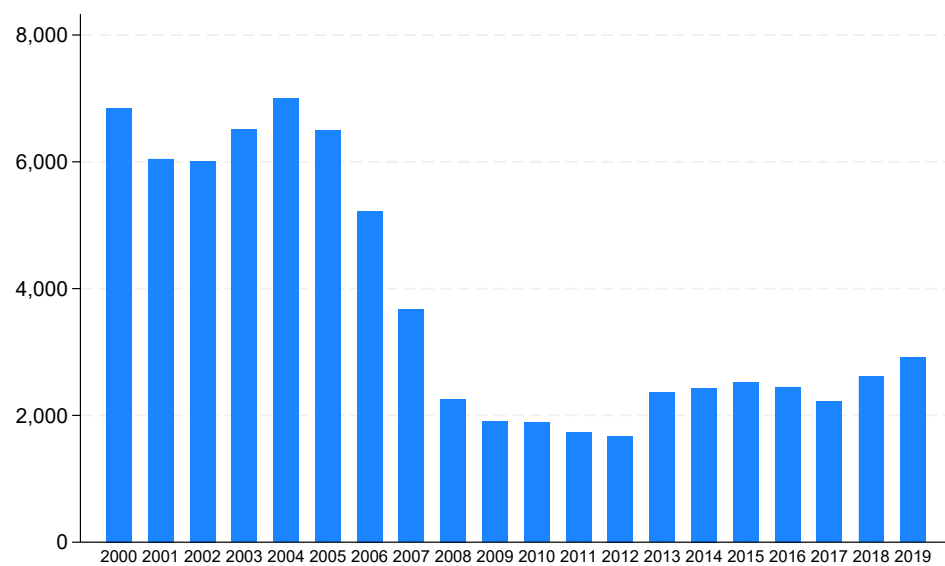


Figure 18: Number of Continuing Entrant Firms Over Time

Notes: This plot shows the number of homes built by entrants in the Cotality built-for-sale data set. This excludes firms who enter and exit in the same year.

6 How Do Different Segments Build?

Different builder size segments build different kinds of home and target different kinds of buyer. From the institutional setting, large builders build similar houses at scale, and thus benefit from economies of scale, and smaller builders tend to focus on higher-end homes. We can verify this in our data. To measure the square footage of the property, I use the Cotality Property data, which contain property-level data from tax assessments. I match these data to our CoreLogic sample using APN number, APN sequence number, and the FIPS code. Some variables include the living area in square feet, the number of bedrooms, and the number of bathrooms. I perform some basic cleaning to remove outliers.¹⁹ I have 2,387,839 observations for which I observe all values and home characteristics. I do not apply these filters for information about the financing of homes. I compute real price per square foot of living area (ppsf) and the real price per bedroom (price_per_bedroom). Table 6 shows summary statistics of these data.

	Mean	50th percentile	Standard deviation	Minimum value	Maximum value
living_area	2405.636	2255	833.1537	990	6187
number_bedrooms	3.578267	4	.8008963	1	6
number_bathrooms	2.911179	3	.8048921	1	6
sale_amount_real	239346.1	206670.1	127461.2	31569.56	1233250
ppsf	100.5856	90.89122	42.58002	7.807833	908.7827
price_per_bedroom	68099.6	58764.17	36349.64	6848.796	1152293

Table 6: Building Facts Summary Statistics

Notes: This table shows descriptive statistics for the variables examined in this section. These data use the Cotality new construction data as the base, but keep observations that are not missing living area, the number of bedrooms, the number of bathrooms, and the price per square foot.

Large builders offer deals relative to the observable characteristics of the house. The mean real sales price per square foot and mean real sales price per bedroom by market segment over time are plotted in Figure 19 and Figure 20. Real prices rose for all builder sizes during the boom, and then fell to levels last seen in 2000 or earlier. Real prices then grew steadily from 2012 onward. In all time periods, large builders are cheaper per measurable unit than small builders, and cheaper than or comparable to medium builders.²⁰ Since these effects could be related to locations, Figure 21 and Figure 22 show the mean of the residuals after accounting for CBSA-year fixed effects. After controlling for CBSA-year fixed effects, homes built by large builders are still relatively cheaper than those built by small firms. This

¹⁹I trim the top and bottom .5% of living area within each year. I also remove the bedroom and bathroom information for homes with values greater than the 99.5th percentile of values within each year.

²⁰This finding is robust to different distributions of prices. In the Appendix, Figure 51, Figure 52, and Figure 53 show the 25th, 50th, and 75th percentiles of the price-per-square-foot distribution. Figure 54, Figure 55, and Figure 56 show the same percentiles for the price per bedrooms.

suggests that large firms may have economies of scale that they pass onto the consumer, or that small firms may provide a luxury product.

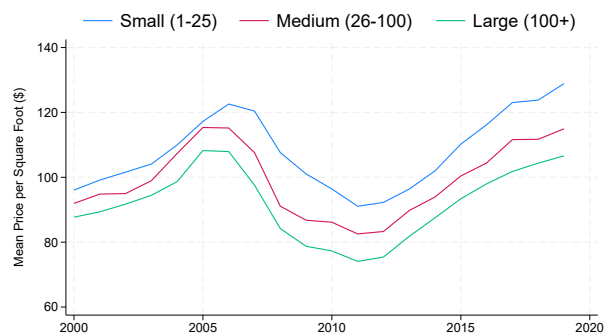


Figure 19: Mean Price per Square Foot by Firm Size

Notes: This plot shows the mean price per square foot by firm size category over time.

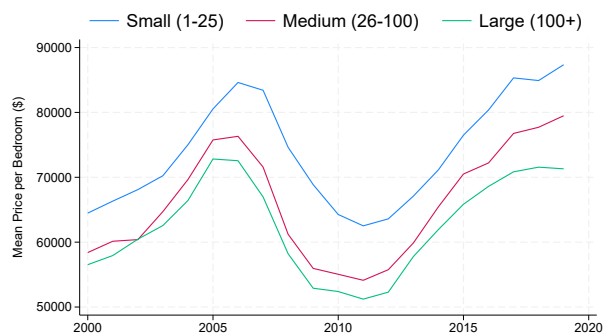


Figure 20: Mean Price per Bedroom by Firm Size

Notes: This plot shows the mean price per bedroom by firm size category over time.

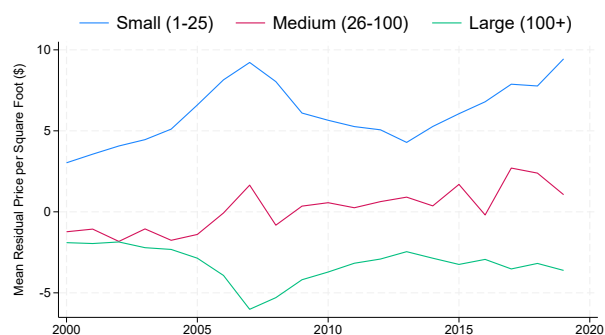


Figure 21: Mean Residual Price per Square Foot by Firm Size

Notes: This plot shows the mean residual price per square foot (location-adjusted) by firm size category over time.

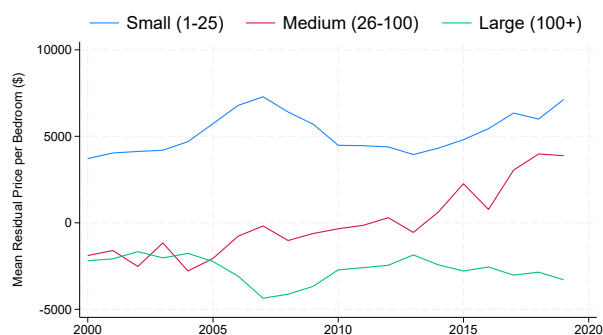


Figure 22: Mean Residual Price per Bedroom by Firm Size

Notes: This plot shows the mean residual price per bedroom (location-adjusted) by firm size category over time.

These firms also build different kinds of structures. Small builders might build a variety of products based on local market trends while large firms offer homes from a set of standard plans. In Figure 24, large firms and small firms offer similar sizes before the crash, but after the crash the average size of homes built by large builders is larger. The size of homes built by large builders rises until 2015 and then falls. After residualizing for CBSA-year fixed effects, small firms seem to offer slightly larger living areas, but this falls throughout the time frame, while large firms offer a more consistent size relative to their CBSA. In areas where large firms build, they may define the local average. Figure 25 shows that the standard deviation of living area for large firms is always lower than that of small and medium firms, but standard deviation for all declines from 2015-2019. Taken together with the fact that large firms generally offer more living space than small and medium firms, this suggests that large builders produce more standardized housing, with a higher living area but lower dispersion within their products and operating markets.

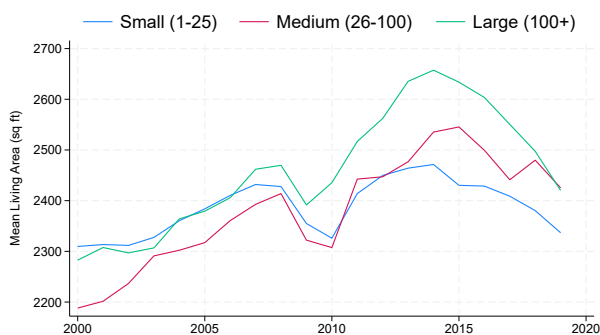


Figure 23: Mean Living Area by Firm Size
Notes: This plot shows the mean living area by firm size category over time.

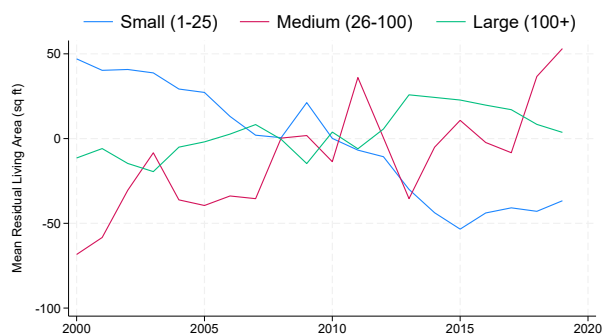


Figure 24: Mean Living Area by Firm Size
Notes: This plot shows the mean living area by firm size category over time.

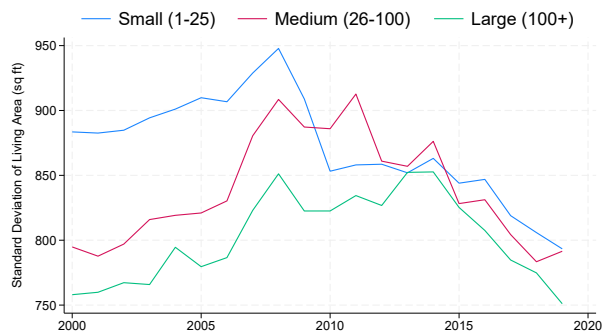


Figure 25: Standard Deviation of Living Area by Firm Size
Notes: This plot shows the standard deviation of living area by firm size category over time.

A consideration for a speculative builder is how the buyer will fund the purchase. Not

all forms of payment are the same. Securing a mortgage can take time and might not happen at all, as banks may delay or deny the mortgage applications of lower-income buyers. These delays are costly, particularly for small speculative builders who have a short-term construction loan deadline. Cash purchases are then associated with lower frictions and less time to sell because there is no need for the customer to acquire a mortgage. As discussed, many large builders have internal mortgage origination units that allow them to provide mortgage financing and earn extra money from selling the mortgages off. Medium and small builders may lack such units due to the high overhead and their lower volumes. Because of these features, we should expect small builders to prefer cash, and large builders to take more mortgage payments. Figure 26 shows that small builders rely on customers making cash purchases for about 10% of their total business. Large builders have a cash purchase share of around 4-5%. There was a dip in the use of cash purchases in all segments during the peak time of the boom. An increased uptake of mortgages during the period is consistent with the increase in credit supply due to loosened credit constraints as in [Justiniano, Primiceri, and Tambalotti \(2019\)](#).

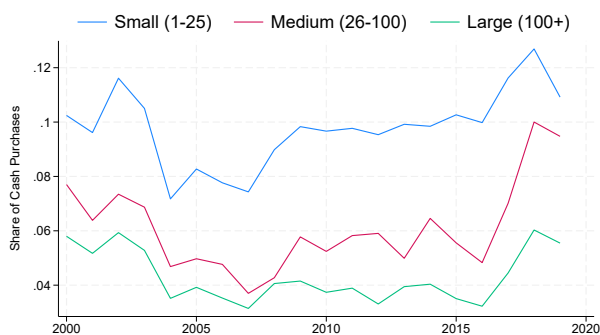


Figure 26: Share of Cash Purchases by Firm Size

Notes: This plot shows the share of cash purchases by firm size category over time.

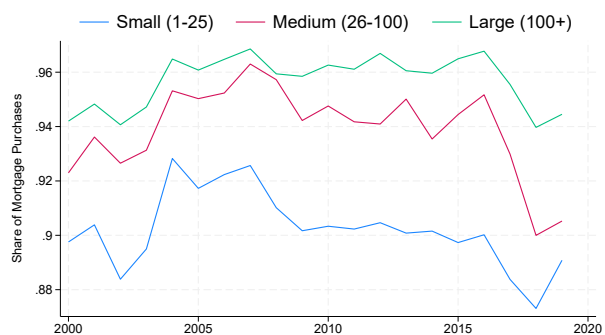


Figure 27: Share of Mortgage Purchases by Firm Size

Notes: This plot shows the share of mortgage purchases by firm size category over time.

Mortgages differ in terms of attractiveness not just to the buyer, but the seller. There are two prominent types of mortgage: the conventional and the Federal Housing Administration (FHA) loan. The conventional mortgage is issued by banks and then either held as part of the portfolio or sold to be securitized. Conventional mortgages typically have a set 20% down payment, and offer lower rates for higher levels of income. Conventional mortgages have much higher rates for those with lower incomes, especially after the Great Recession. During the boom, down payment requirements and initial interest rates fell, and mortgage credit became more available. FHA loans are loans intended for first-time and lower-income homebuyers. FHA loans can only be used for a buyer's primary residence, but are more

flexible financially. Guaranteed by the federal government and administered by local banks, FHA loans usually have lower down payment and FICO score requirements, and can offer lower interest rates than conventional loans for buyers with lower credit scores. For builders, a downside of taking FHA loans as payment is that they have an additional inspection and until March 2019, required a ten-year warranty from the builder. After March 2019, the warranty period changed to one year.²¹

Small builders might prefer to avoid additional inspections or lack the resources or institutional knowledge to provide a ten-year warranty. We should expect small builders to have a lower share of FHA mortgages within mortgages and large builders to have a higher share of FHA mortgages. Figure 28 and Figure 29 show the share of purchases funded by FHA and conventional mortgages respectively. In all periods, large builders have a higher FHA share than small builders. Before the boom, 20% of large builder business in FHA loans. During the housing boom, there were lower down payment requirements and looser restrictions for conventional mortgages, so all segments accepted more conventional mortgages as payment. When access to conventional mortgage credit dried up in the bust, all segments began accepting FHA loans at higher rates than previously, but large builders continued to have higher FHA shares than other builder sizes.

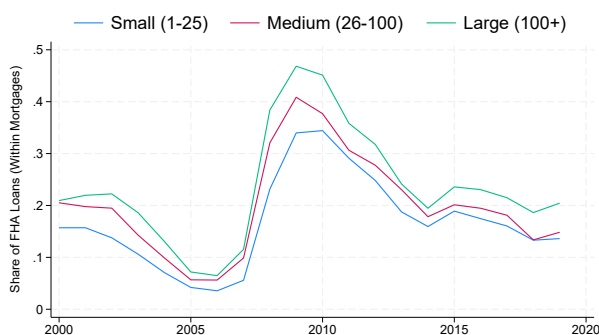


Figure 28: Share of FHA Loans by Firm Size Within Mortgages

Notes: This plot shows the share of FHA loans by firm size category over time, conditional on the purchase being financed by a mortgage.

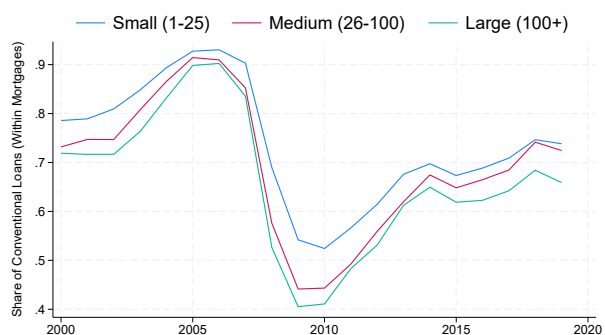


Figure 29: Share of Conventional Loans by Firm Size Within Mortgages

Notes: This plot shows the share of conventional loans by firm size category over time, conditional on the purchase being financed by a mortgage.

7 Conclusion

The traditional driver of new housing supply, speculative building, was disrupted by the housing bust. Construction, in the aggregate and cross-section, fell by 75% and never recov-

²¹See [Federal Register](#).

ered to its 2000 output, while prices recovered within three years of hitting their low in 2012. The number of firms fell by over half during the bust. Firm size heterogeneity separates these different trajectories. Small and medium builders produce fewer homes than they did in 2000, and entry is low. Large builders now exceed their 2000 production of new homes and the recovery in new homes is driven by the *largest* firms. Because of the way builders finance and do business, firm segments behave differently in what products they offer, what forms of payment they transact with, and their relative recoveries after the crash. These facts are consistent with tighter credit availability for firms after the crisis, which stifled smaller firm activity and contributed materially to the housing shortage.

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A Recovery Plots

A.1 General Facts

	Total Units	Built for Sale	Built for Rent	Owner Built	Contractor Built
year					
1999	1.01622	.9952596	1.034119	1.08962	1.059994
2000	1	1	1	1	1
2001	1.000744	1.01464	.9391039	.9602649	.9769859
2002	1.054013	1.081526	.9715269	.9634073	1.005405
2003	1.114424	1.173687	1.086437	.9338927	.9679372
2004	1.231826	1.32345	1.124312	.9851916	.9976456
2005	1.304329	1.445791	.9555092	.9207262	.9838361
2006	1.321048	1.453693	.9610827	.9666517	1.024085
2007	.9812229	1.019935	.9639421	.8522699	.8926166
2008	.666739	.6296505	1.221363	.7423975	.6679071
2009	.41842	.3728505	.7458987	.519298	.4905355
2010	.3974472	.3714188	.640221	.4161659	.452423
2011	.3556206	.3255883	.7632006	.3799738	.3898931
2012	.3854059	.3668155	.7641457	.3685749	.4006102
2013	.457189	.4676912	.8018029	.328502	.4202251
2014	.4957369	.5064208	.6629995	.3402072	.5137367
2015	.5198966	.5382341	.6945986	.3548585	.5074739
2016	.5916106	.6224249	.8061405	.3620466	.5560851
2017	.6366475	.6758197	.957254	.3657075	.5673947
2018	.6729796	.7170163	1.108367	.39893	.5588449
2019	.7258046	.7629523	1.259481	.4137384	.6473514
2020	.7267132	.7752705	1.222284	.4161659	.6030748
2021	.7789569	.8621221	1.268544	.3445673	.5797384
2022	.8298919	.8835722	1.661497	.4064874	.6853392
2023	.8035742	.7971227	2.368987	.4545812	.7283165

Table 7: SOC Relative Recovery by Market Segment

Notes: The table shows the number of new units produced in each market segment of the Survey of Construction normalized to 2000 levels of production.

A.2 Permits Analysis

	Mean
year	
2000	1
2005	.9649924
2010	.304414
2015	.3957382
2019	.4901065

(a) CBSA Recovery (Unweighted)

	Mean
year	
2000	1
2005	.9872223
2010	.1651281
2015	.3137627
2019	.4613504

(b) CBSA Recovery (Population-Weighted)

Table 8: CBSA Recovery of Permits: Comparison of Unweighted vs. Population-Weighted
Notes: These tables compute the share of CBSAs in the data that match or exceed their lowest level of 1990s single-family housing production. In the first table, all CBSAs are equally weighted, so this represents the share of CBSAs that recovered. The table on the right is weighted by population, so it instead gives us the share of people who live in a CBSA that is producing as much housing in that year as it did during the lowest year of the 1990s.

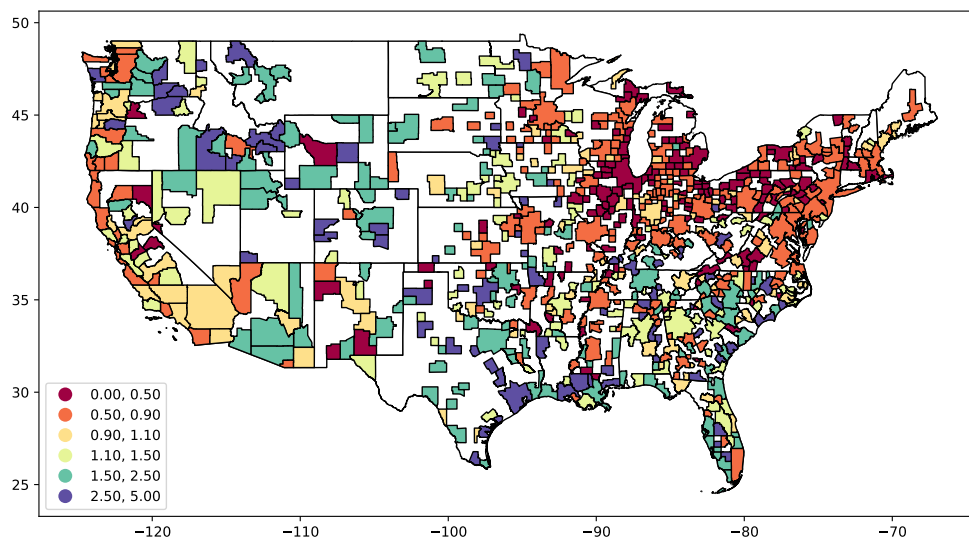


Figure 30: Ratio for Single-Family Permits Relative to Minimal 1990s Levels

Notes: This plot is a heat-map of the ratio of 2019 housing permits rate (new single-family permits per thousand new households) to the lowest level of permits rate during the 1990s.

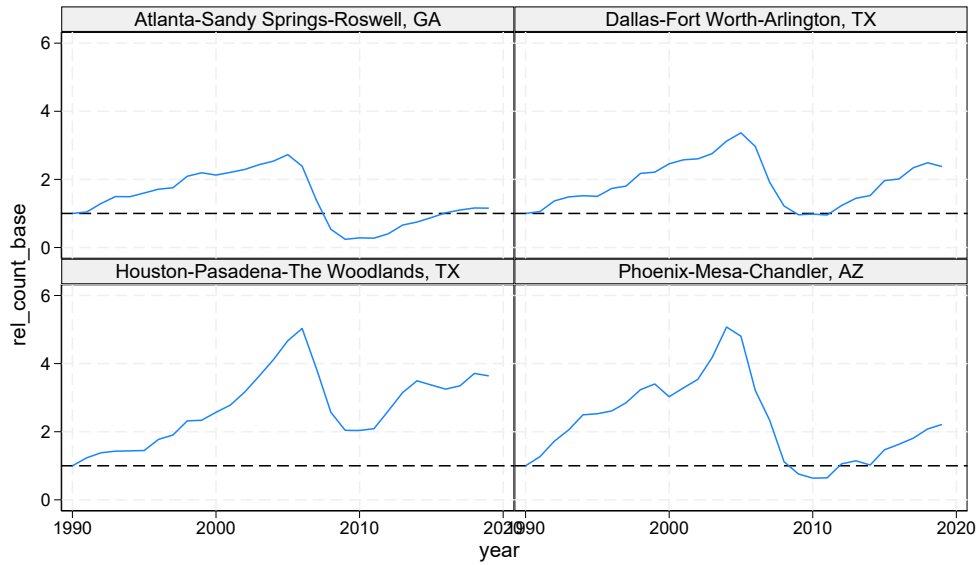


Figure 31: High Construction CBSA Permits

Notes: This plot shows the number of new single-family permits over time for several low-regulation, high-production housing markets.

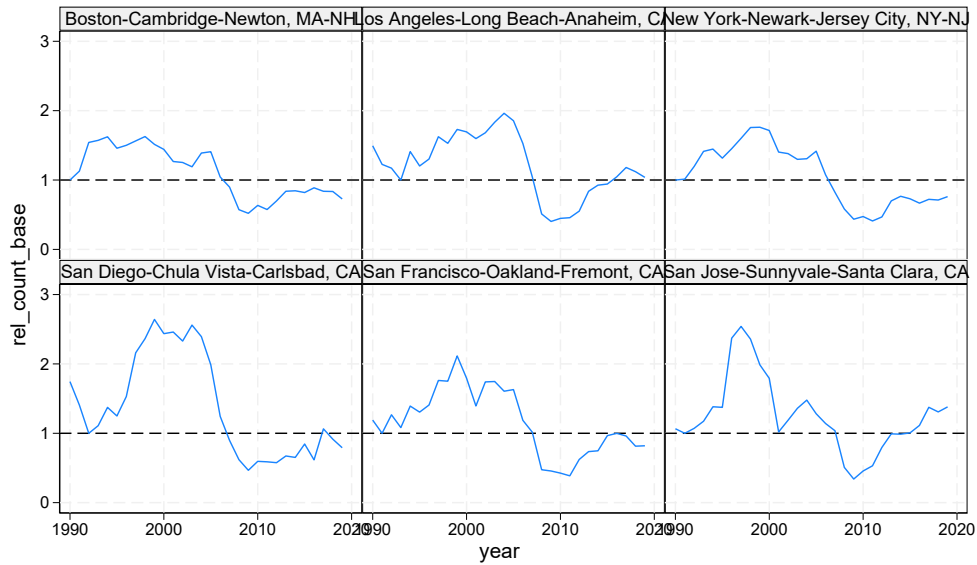


Figure 32: Low-Production CBSA Permits

Notes: This plot shows the number of new single-family permits over time for several major cities with historically high regulation levels.

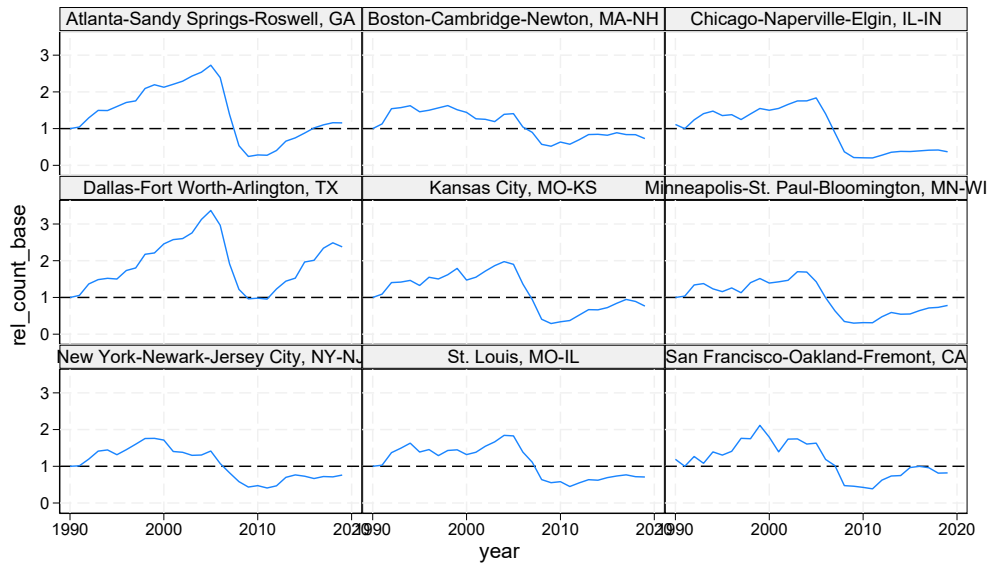


Figure 33: Federal Reserve CBSA Permits Rate

Notes: This plot shows the number of new single-family permits over time for several major cities spread throughout the country that house regional Federal Reserve Banks.

A.3 Permits Rate Supplemental Tables

	Mean	10th percentile	25th percentile	50th percentile	75th percentile	90th percentile
year						
1990	153.6336	44.18985	88.69908	147.3397	203.1839	273.2517
2000	192.5414	73.25194	119.3269	183.6228	256.7744	328.0721
2010	84.84307	30.92395	48.583	75.71425	112.3755	151.9952
2019	111.6206	39.18449	59.38242	94.21909	151.9195	210.3583

Table 9: Permits Rate Distribution

Notes: This plot shows the distribution of new single-family home production divided by the number of new households in thousands for the years 1990, 2000, 2010, and 2019. Each distribution is trimmed at the 1st and 99th percentiles within that year. Permits data come from the Building Permits Survey and inflow data come from the SOI Migration Statistics.

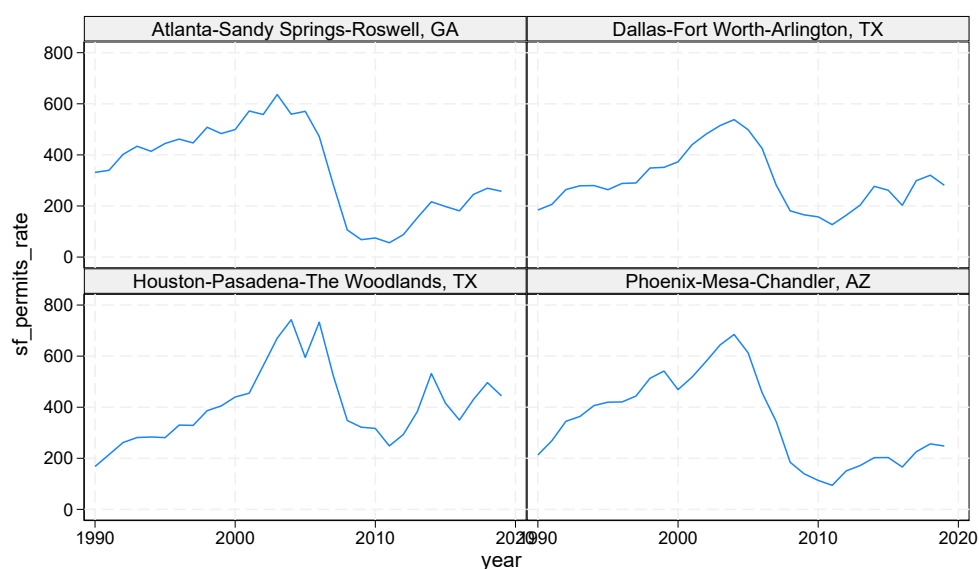


Figure 34: High Construction CBSA Permits Rate

Notes: This plot shows the ratio of new single-family permits to thousand new potential households over time for several low-regulation, high-production housing markets. Permits are taken from the BPS and new inflows are taken from SOI migration data.

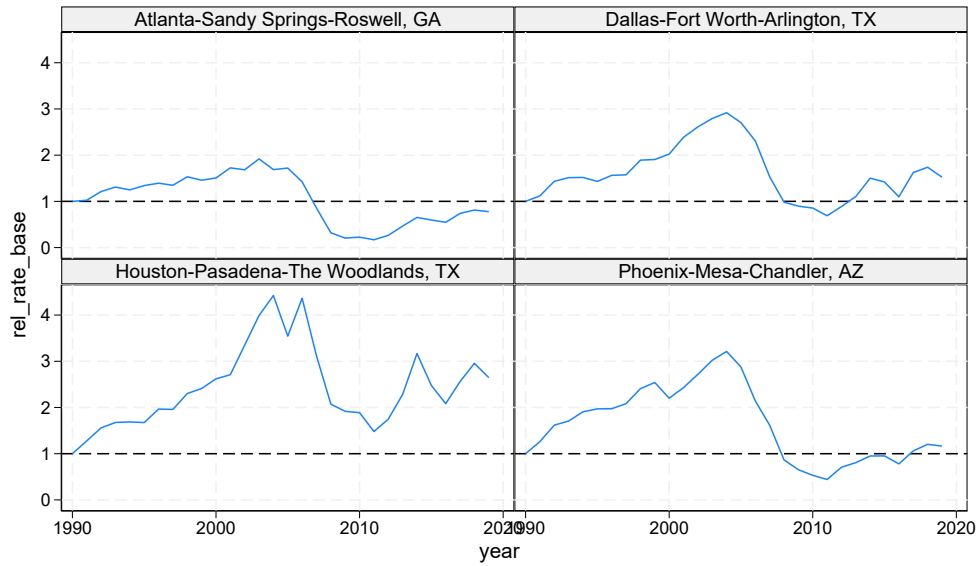


Figure 35: High Construction CBSA Relative Permits Rate

Notes: This shows the ratio of new single-family permits to thousand new potential households over time for several low-regulation, high-production housing markets. Values are normalized by the lowest value of the ratio in the 1990s. Permits are taken from the BPS and new inflows are taken from SOI migration data.

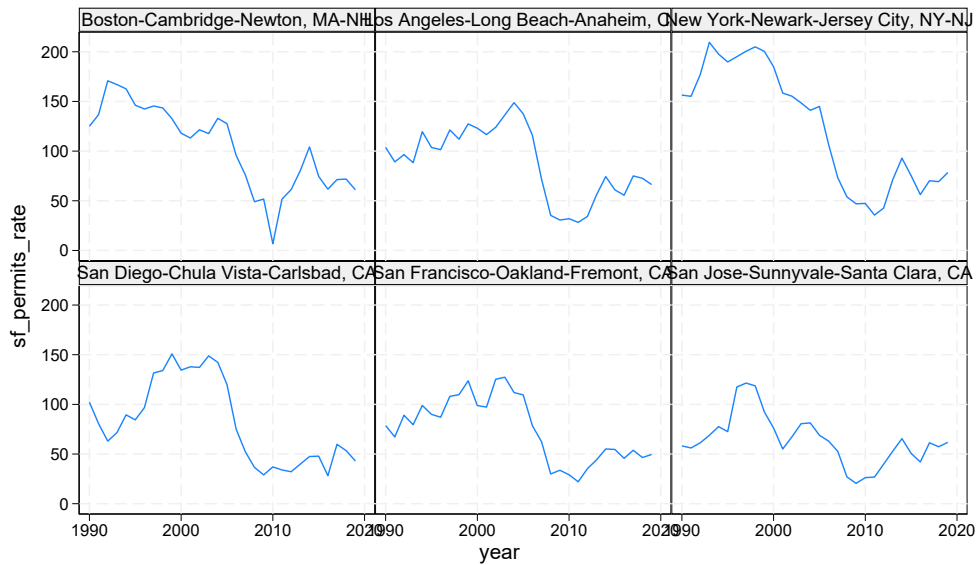


Figure 36: Low-Production CBSA Permits Rate

Notes: This plot shows the ratio of new single-family permits to thousand new potential households over time for several major cities with historically high regulation levels. Permits are taken from the BPS and new inflows are taken from SOI migration data.

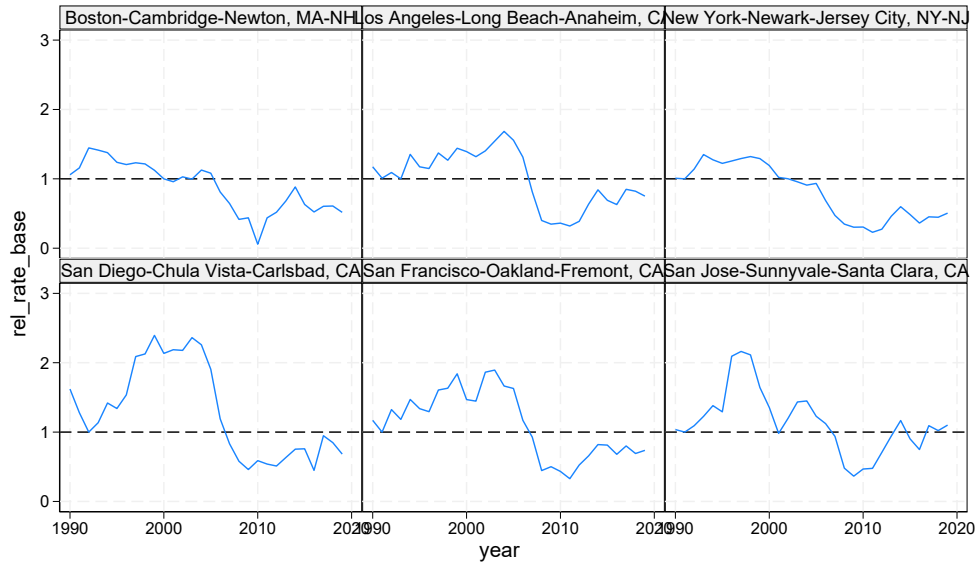


Figure 37: Low-Production CBSA Relative Permits Rate

Notes: This plot shows the ratio of new single-family permits to thousand new potential households over time for several major cities with historically high regulation levels. Values are normalized by the lowest value of the ratio in the 1990s. Permits are taken from the BPS and new inflows are taken from SOI migration data.

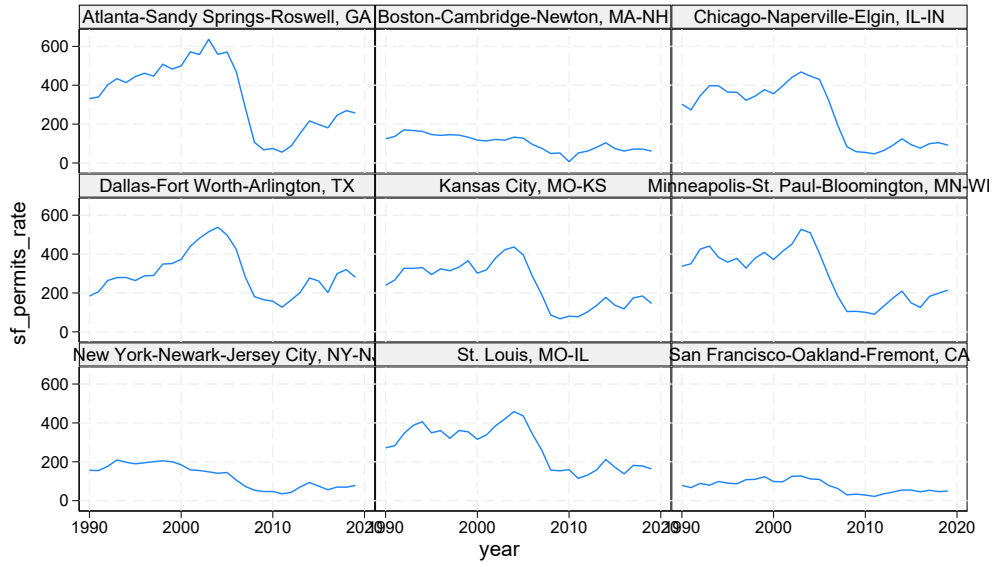


Figure 38: Federal Reserve CBSA Permits Rate

Notes: This plot shows the ratio of new single-family permits to thousand new potential households over time for several major cities spread throughout the country that house regional Federal Reserve Banks. Permits are taken from the BPS and new inflows are taken from SOI migration data.

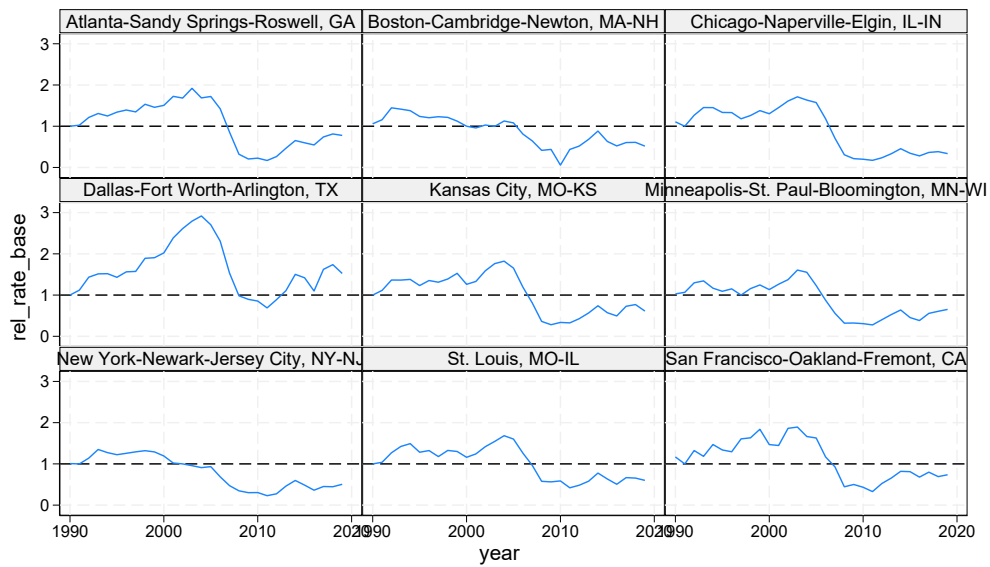


Figure 39: Federal Reserve CBSA Relative Permits Rate

Notes: This plot shows the ratio of new single-family permits to thousand new potential households over time for several major cities spread throughout the country that house regional Federal Reserve Banks. Values are normalized by the lowest value of the ratio in the 1990s. Permits are taken from the BPS and new inflows are taken from SOI migration data.

B Large Firm Appendix Plots

B.1 Aggregate Supplemental Figures

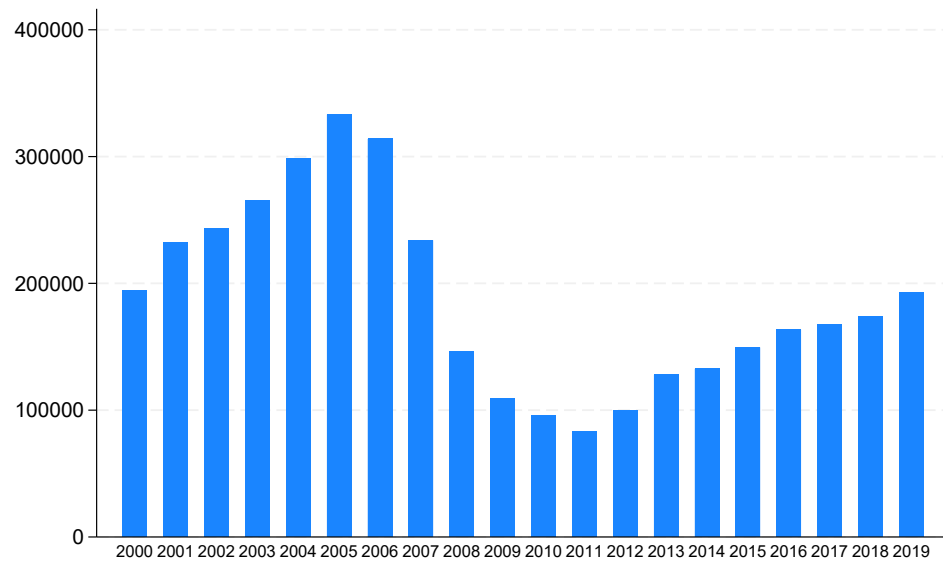


Figure 40: Total Number of Homes Produced Over Time

Notes: This plot shows the total number of homes produced over time from the Cotality built-for-sale data set.

B.2 Firm Descriptive Statistics

	Mean	P25	Median	P75	P90	Total	Firm Count
year							
2000	2.764865	1	1	3	7	77842	28154
2001	2.815893	1	1	3	7	85330	30303
2002	2.828109	1	1	3	7	85802	30339
2003	2.874426	1	1	3	7	90806	31591
2004	2.90386	1	1	3	7	102091	35157
2005	2.893493	1	1	3	7	107881	37284
2006	2.828849	1	1	3	7	98724	34899
2007	2.647672	1	1	3	6	83354	31482
2008	2.47893	1	1	2	5	57650	23256
2009	2.477522	1	1	2	5	44915	18129
2010	2.496034	1	1	2	6	39023	15634
2011	2.492096	1	1	2	6	32948	13221
2012	2.698338	1	1	3	6	36209	13419
2013	2.781184	1	1	3	7	43634	15689
2014	2.823852	1	1	3	7	45769	16208
2015	2.822011	1	1	3	7	48643	17237
2016	2.766448	1	1	3	6	51171	18497
2017	2.695999	1	1	3	6	48315	17921
2018	2.479838	1	1	2	5	50182	20236
2019	2.582609	1	1	2	6	52600	20367

Table 10: Home Production for Small Builders

Notes: This table shows summary statistics for the home production of small builders (1-25 homes) in the Cotality new construction database over time. It also shows the number of firms active.

	Mean	P25	Median	P75	P90	Total	Firm Count
year							
2000	46.52995	31	40	58	78	30291	651
2001	46.17958	31	40	56	78	37544	813
2002	47.33454	31	41	58	78	39335	831
2003	46.98249	31	40	56	78	42942	914
2004	45.92048	30	41	56	75	46196	1006
2005	45.60378	31	39	57	76	50757	1113
2006	46.43369	32	40	56	75	44112	950
2007	46.81555	31	40.5	57	80	30711	656
2008	45.6399	31	39	56	76	17617	386
2009	46.34722	31	40	56	79	13348	288
2010	46.92391	32	41	60	74	12951	276
2011	46.11428	31	41	55	76	11298	245
2012	45.52337	31	39	55	75	14613	321
2013	45.66979	31	38	57	77	19501	427
2014	46.27273	32	40	56	80	18833	407
2015	45.30679	31	39	56	74	19346	427
2016	45.62719	30	39	57	77	20806	456
2017	46.33188	31	39	56	76	21220	458
2018	47.14824	31	41	59	79	20038	425
2019	47.38334	31	41	60	80	22744	480

Table 11: Home Production for Medium Builders

Notes: This table shows summary statistics for the home production of medium builders (26-100 homes) in the Cotality new construction database over time. It also shows the number of firms active.

	Mean	P25	Median	P75	P90	Total	Firm Count
year							
2000	452.2094	127	199	371	979	86372	191
2001	482.5771	129	177	368	838	109545	227
2002	507.6566	130	187	355	993	118284	233
2003	514.7608	132	188	387	975	131264	255
2004	535.847	129	178	378	1133	150573	281
2005	570.1961	124	177	316	1033	174480	306
2006	648.1477	136	195	426	1112	171111	264
2007	651.5978	139.5	199	398.5	1418	119894	184
2008	617.2174	132	205	536	1441	70980	115
2009	561.1539	125	198	541	1278	51065	91
2010	549.75	133.5	190	583	1480.5	43980	80
2011	494.5316	126	160	500	1327	39068	79
2012	539.0659	136	196	496	1531	49055	91
2013	553.9231	126	202	401	1604	64809	117
2014	566.9083	127.5	204	441.5	1537	68029	120
2015	569.8042	135	206	422	1146	81482	143
2016	589.671	134	206	430	958	91399	155
2017	605.4568	133	208.5	426	1004	98084	162
2018	615.6964	143	207	430.5	1069	103437	168
2019	656.1788	136	213	391	1265	117456	179

Table 12: Home Production for Large Builders

Notes: This table shows summary statistics for the home production of large builders (more than 100 homes) in the Cotality new construction database over time. It also shows the number of firms active.

B.3 Size Category Supplemental Figures

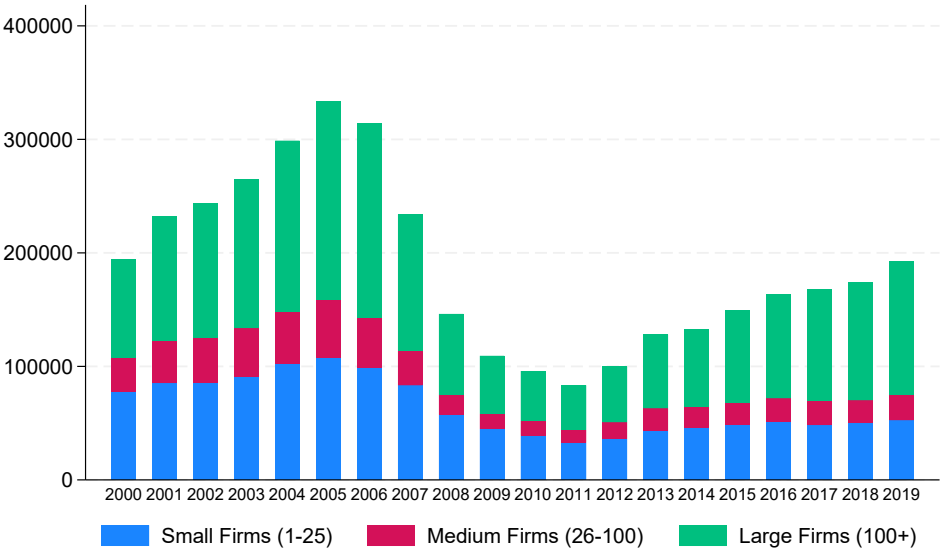


Figure 41: Home Production by Size Category Over Time

Notes: This plot shows the total number of homes produced over time by NAHB size category from the Cotality built-for-sale data set.

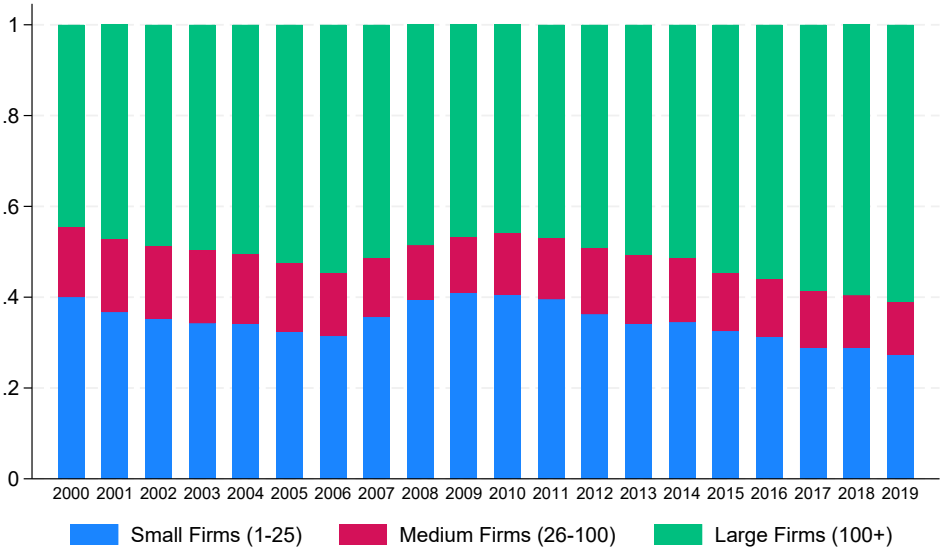


Figure 42: Home Market Shares by Size Category Over Time

Notes: This plot shows the market share of each NAHB size category from the Cotality built-for-sale data set.

	Small Builder (1-25 homes)	Medium Builder (26-100 homes)	Large Builder (>100 homes)
year			
2000	.9709615	.0224514	.0065871
2001	.9668187	.0259388	.0072424
2002	.9661179	.0264624	.0074197
2003	.9643162	.0278999	.0077839
2004	.9646856	.027604	.0077105
2005	.9633362	.0287575	.0079064
2006	.9663833	.0263063	.0073104
2007	.9740115	.0202958	.0056927
2008	.9789115	.0162478	.0048407
2009	.9795223	.0155608	.0049168
2010	.9777361	.0172608	.0050031
2011	.9760798	.0180879	.0058324
2012	.9702119	.0232087	.0065794
2013	.966488	.0263044	.0072075
2014	.9685091	.0243203	.0071706
2015	.9679901	.0239793	.0080305
2016	.9680238	.0238643	.0081118
2017	.9665606	.024702	.0087374
2018	.9715301	.0204042	.0080657
2019	.9686579	.0228289	.0085133

Table 13: Share of Total Firms by NAHB Size Category

Notes: This table shows the share of firms within different NAHB size categories from the Cotality built-for-sale data set.

	Small Builder (1-25 homes)	Medium Builder (26-100 homes)	Large Builder (>100 homes)
year			
2000	.412679	.1605876	.4267333
2001	.3814364	.1678267	.4507369
2002	.3680152	.1687126	.4632722
2003	.3586165	.1695891	.4717944
2004	.3593956	.1626259	.4779785
2005	.3448749	.1622604	.4928647
2006	.3392577	.1515876	.5091546
2007	.3851243	.1418955	.4729802
2008	.4234549	.1294017	.4471434
2009	.4457622	.1324732	.4217646
2010	.4400726	.1460518	.4138756
2011	.4315446	.1479784	.420477
2012	.3991424	.1610834	.4397742
2013	.3770849	.1685276	.4543875
2014	.3888582	.1600071	.4511347
2015	.3695911	.1469916	.4834173
2016	.3558954	.1447062	.4993984
2017	.3297547	.1448286	.5254167
2018	.3325271	.1327803	.5346926
2019	.3209352	.1387709	.5402939

Table 14: Share of Total Homes Built by NAHB Size Category excluding D.R. Horton and Lennar

Notes: This table shows the share of homes built by firms within different NAHB size categories from the Cotality built-for-sale data set. This excludes the two largest home builders, D.R. Horton and Lennar.

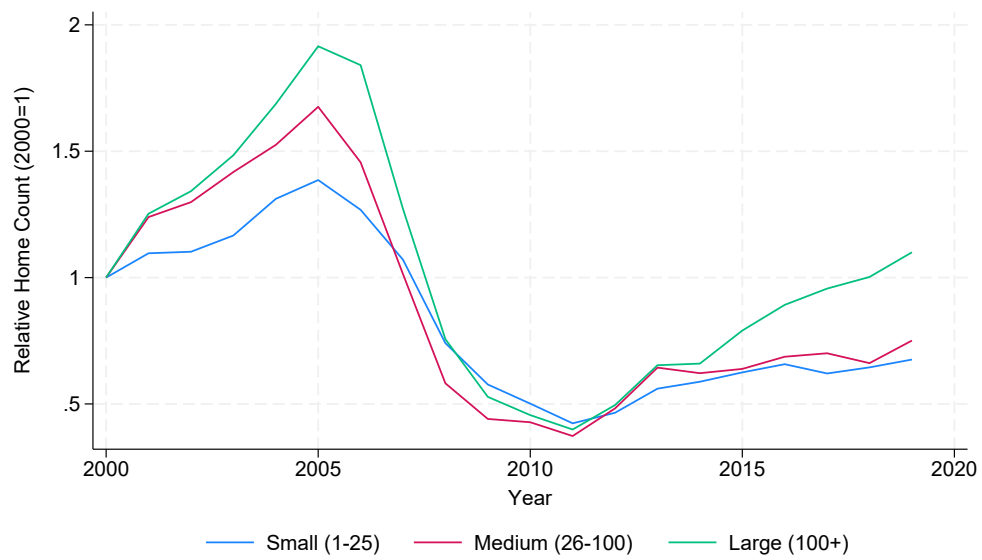


Figure 43: Relative Home Production by Size Category excluding Big Two (2000=1)
Notes: This plot shows the relative number of homes produced by each NAHB size category from the Cotality built-for-sale data set, but excluding totals produced by D.R. Horton and Lennar. The sum is normalized to be 1 in 2000.

B.4 Survival Analysis of Top Builders

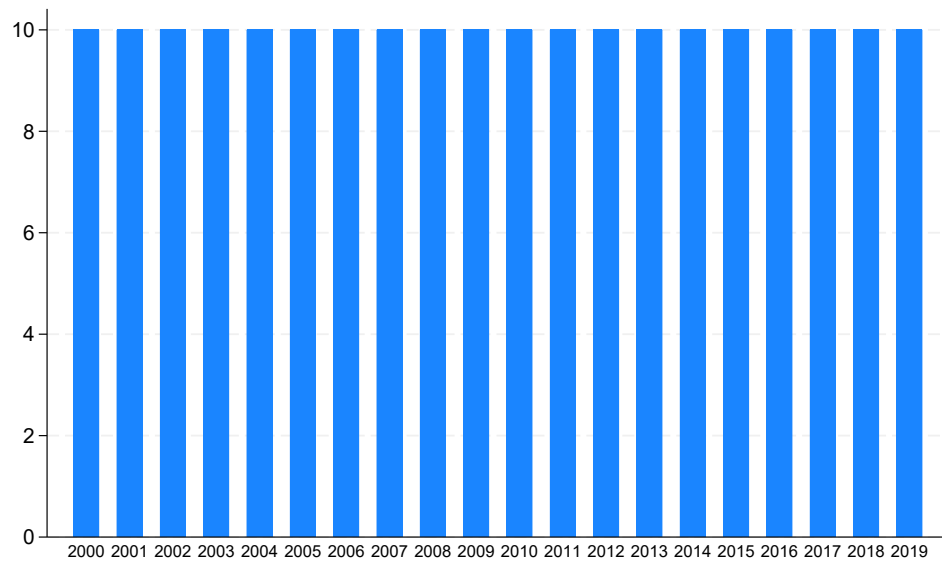


Figure 44: Number of Top 10 Builders from 2000 Surviving Over Time

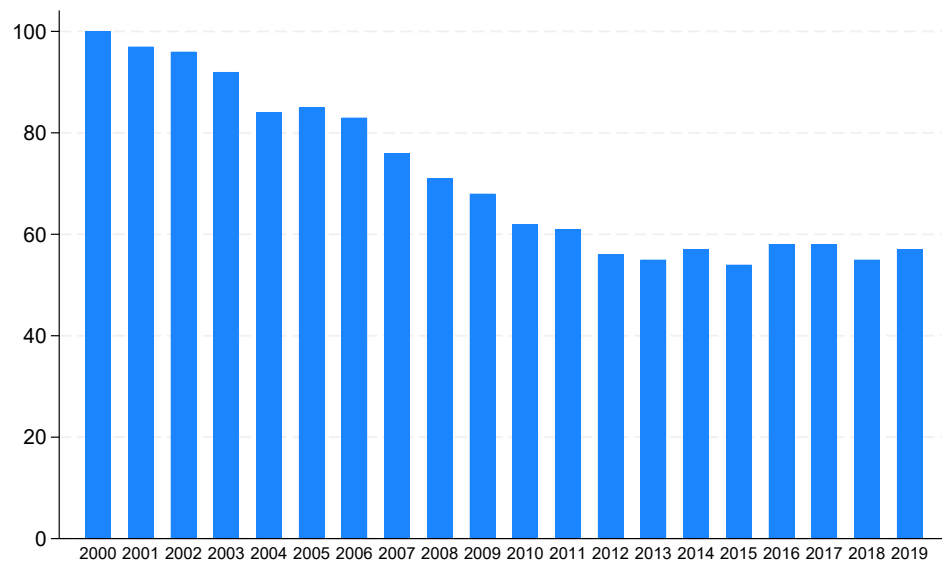


Figure 45: Number of Top 100 Builders from 2000 Surviving Over Time

B.5 Entrant Supplemental Figures

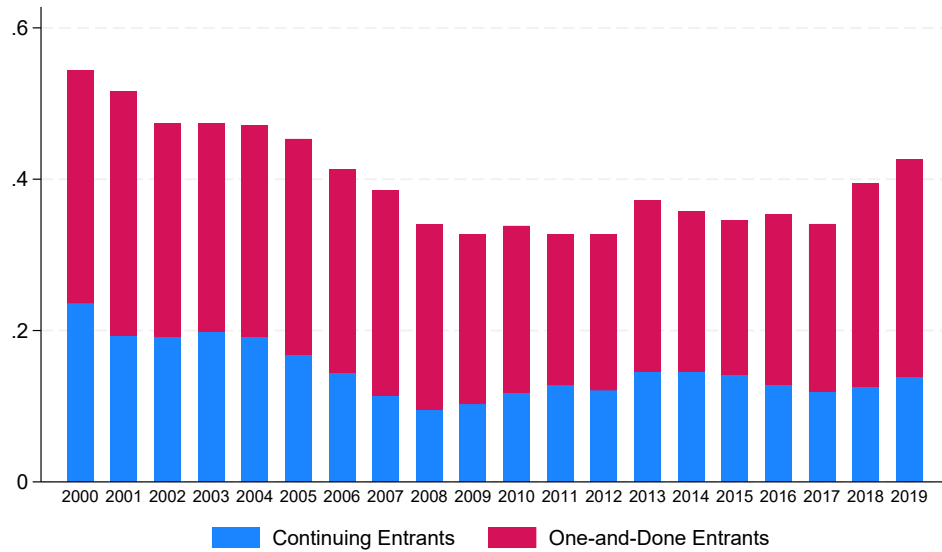


Figure 46: Entry and One-and-Done Firm Shares Over Time

Notes: This plot shows the share of firms who are entrants in the Cotality built-for-sale data set. One-and-Done firms are those who enter and exit in the same year (i.e. this is their only year of operation).

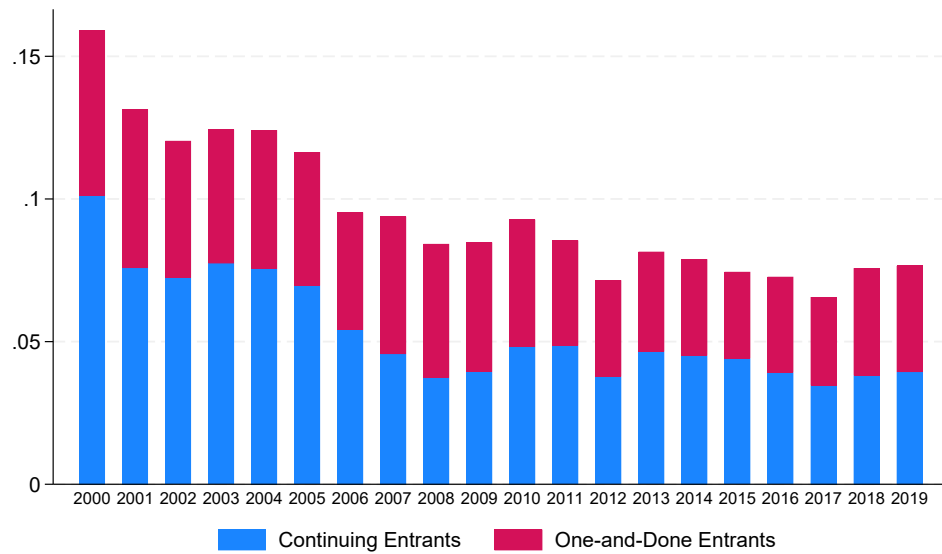


Figure 47: Entry and One-and-Done Home Shares Over Time

Notes: This plot shows the share of homes built by entrants in the Cotality built-for-sale data set. One-and-Done firms are those who enter and exit in the same year (i.e. this is their only year of operation).

B.6 Relative Recovery Analysis by Firm Size Percentiles

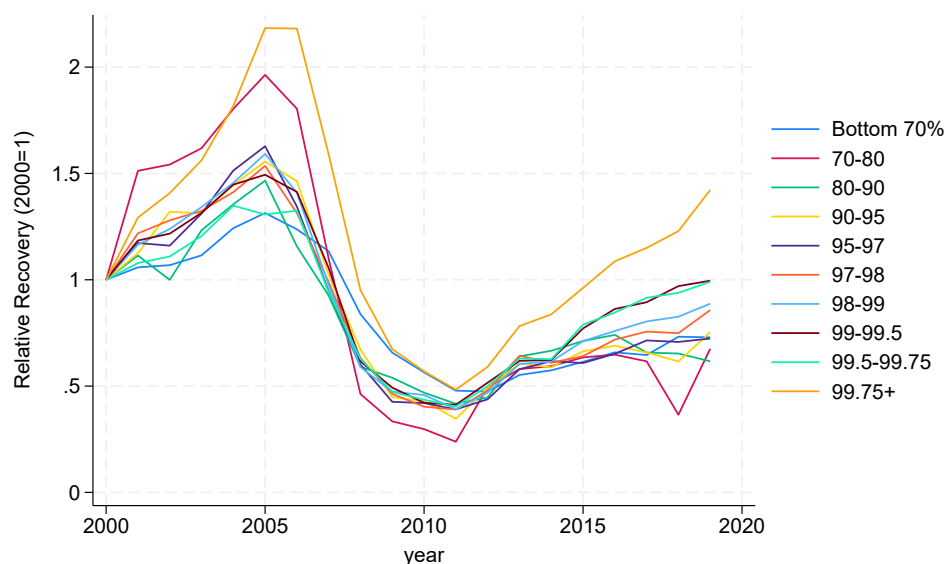


Figure 48: Relative Recovery by Firm Size Percentiles (2000=1)

Notes: The plot shows the production of different percentiles of the firm distribution over time, normalized so each percentile's total production is equal to 1 in 2000.

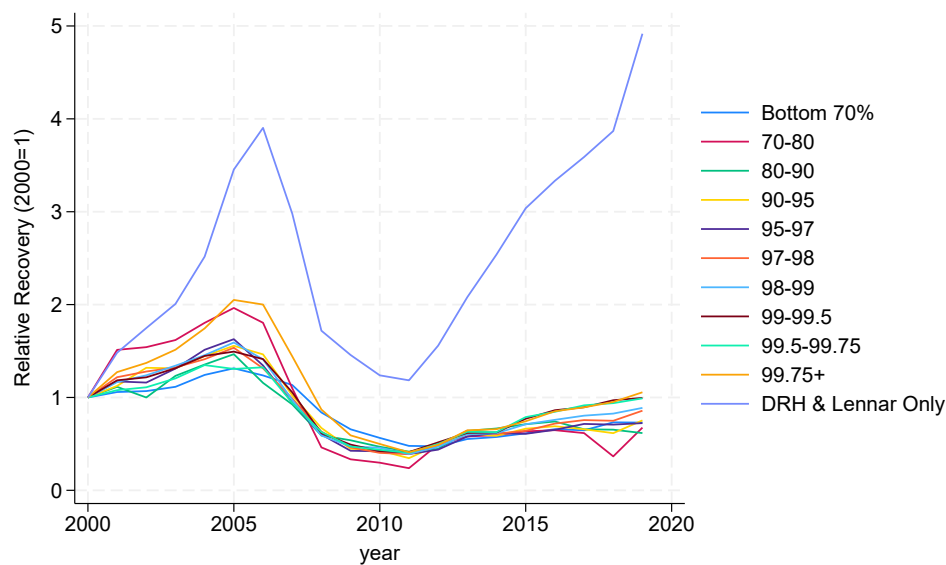


Figure 49: Relative Recovery by Firm Size Percentiles Including DRH & Lennar (2000=1)

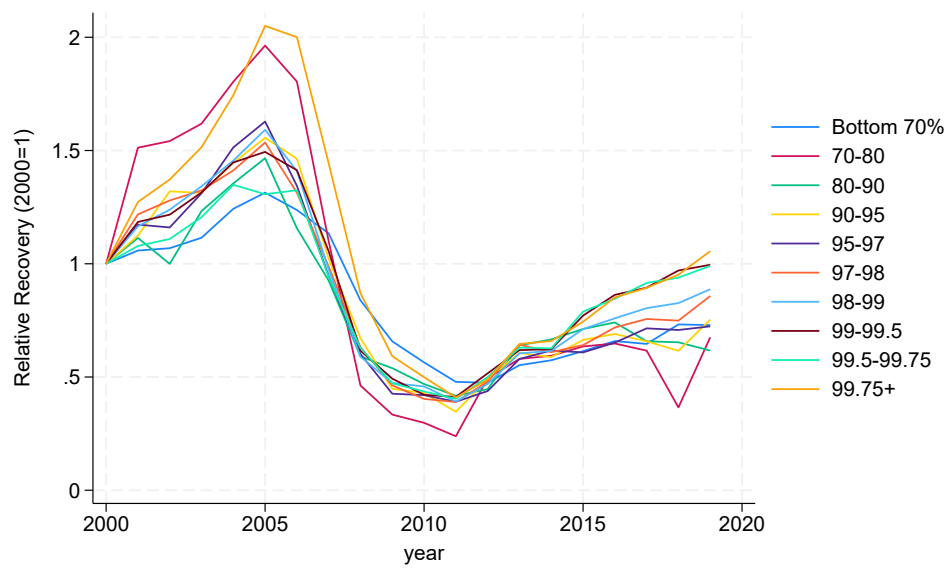


Figure 50: Relative Recovery by Firm Size Percentiles Excluding DRH & Lennar (2000=1)

C Home Characteristics Supplemental Figures

C.1 Pricing Supplemental Figures

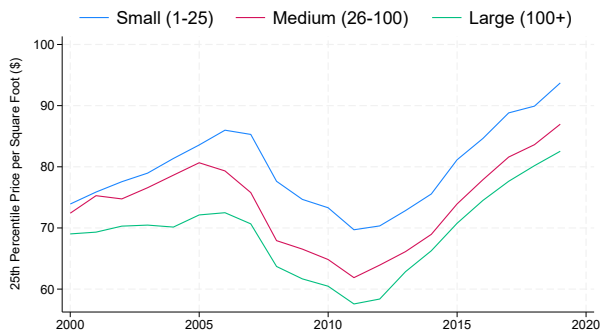


Figure 51: 25th Percentile Price per Square Foot by Firm Size

Notes: This plot shows the 25th percentile price per square foot by firm size category over time.

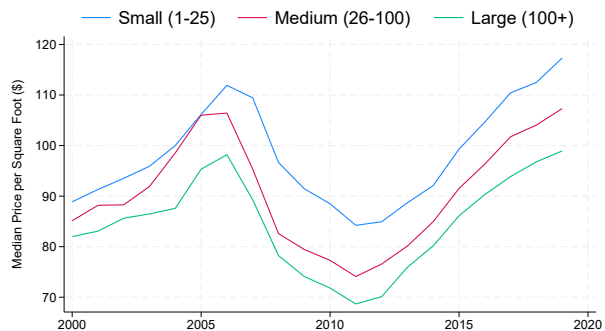


Figure 52: Median Price per Square Foot by Firm Size

Notes: This plot shows the median price per square foot by firm size category over time.

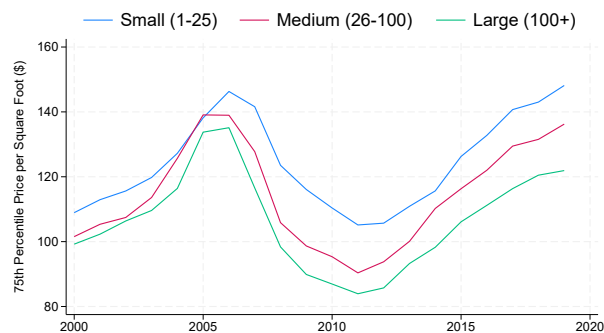


Figure 53: 75th Percentile Price per Square Foot by Firm Size

Notes: This plot shows the 75th percentile price per square foot by firm size category over time.

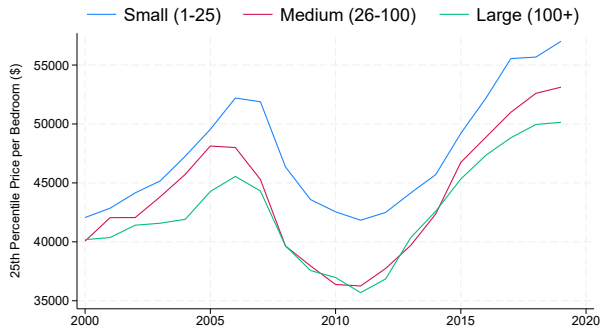


Figure 54: 25th Percentile Price per Bedroom by Firm Size

Notes: This plot shows the 25th percentile price per bedroom by firm size category over time.

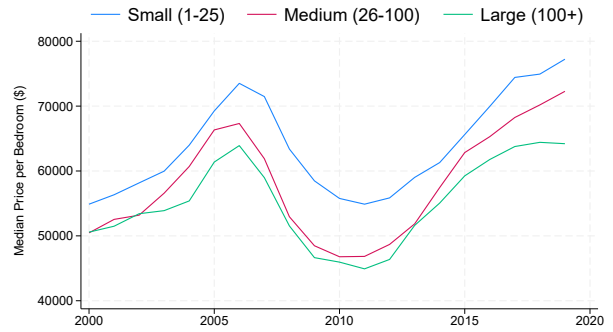


Figure 55: Median Price per Bedroom by Firm Size

Notes: This plot shows the median price per bedroom by firm size category over time.

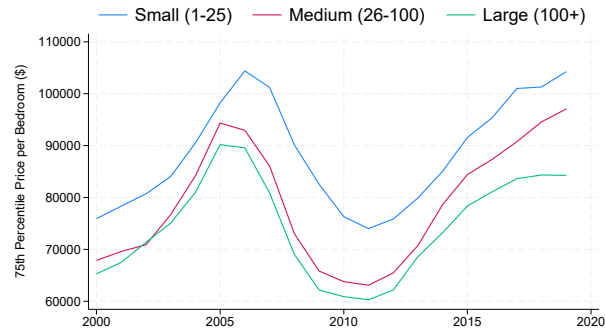


Figure 56: 75th Percentile Price per Bedroom by Firm Size

Notes: This plot shows the 75th percentile price per bedroom by firm size category over time.

C.2 Physical Characteristics Supplemental Figures

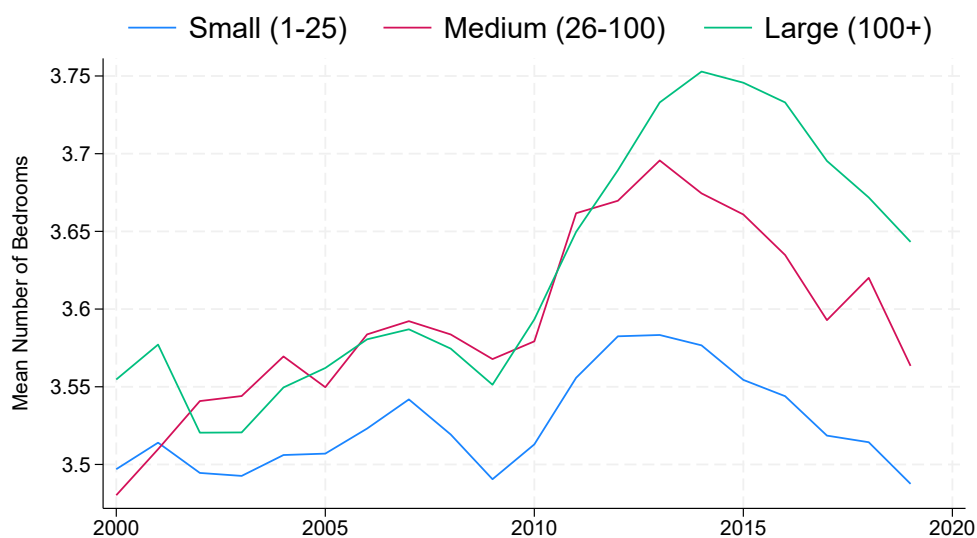


Figure 57: Mean Number of Bedrooms by Firm Size

Notes: This plot shows the mean number of bedrooms by firm size category over time.

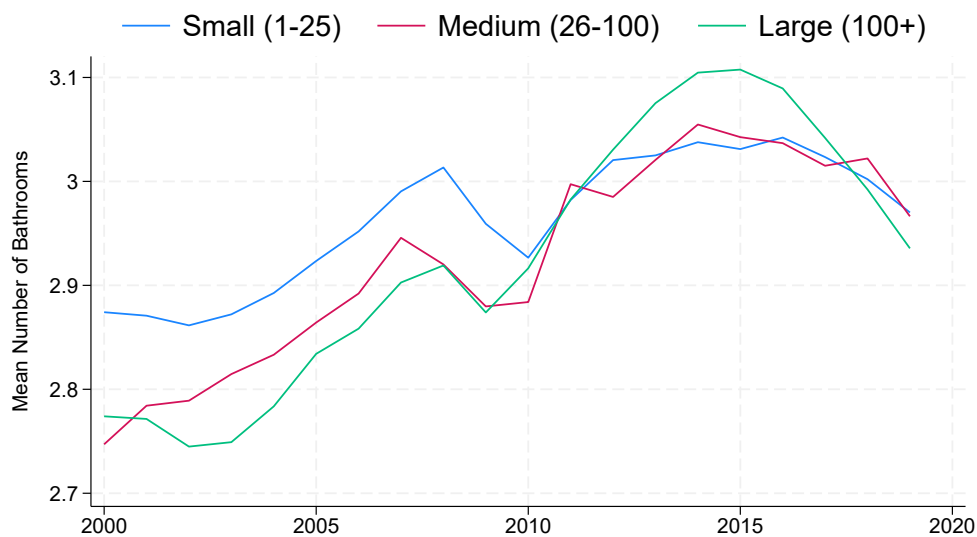


Figure 58: Mean Number of Bathrooms by Firm Size

Notes: This plot shows the mean number of bathrooms by firm size category over time.

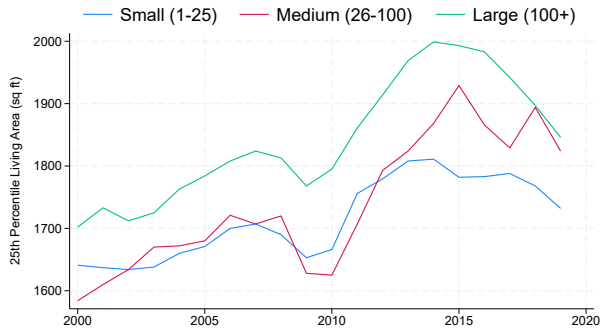


Figure 59: 25th Percentile Living Area by Firm Size

Notes: This plot shows the median living area by firm size category over time.

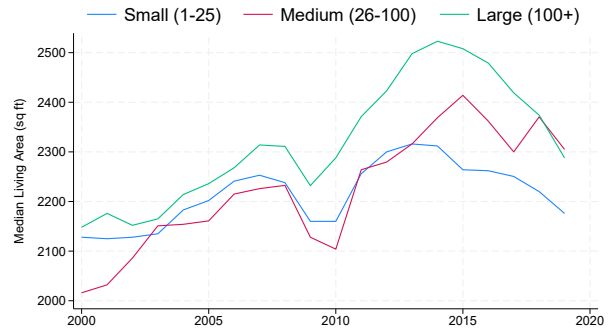


Figure 60: Median Living Area by Firm Size

Notes: This plot shows the median living area by firm size category over time.

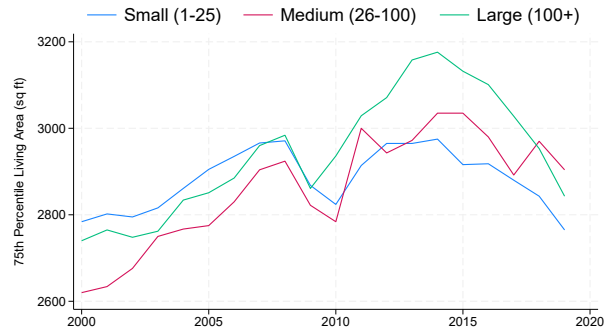


Figure 61: 75th Percentile Living Area by Firm Size

Notes: This plot shows the 75th percentile living area by firm size category over time.